T-FDR: Trust-guided frequency-aware diffusion refinement for reliable cardiac MRI segmentation under limited annotations

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Abstract

Semi-supervised image segmentation is often limited by noisy pseudo-label propagation and insufficient modeling of boundary-sensitive features and latent semantic distributions, especially in cardiac magnetic resonance imaging (MRI) with low contrast and ambiguous structures. To address these challenges, we propose a trust-guided frequency-aware diffusion refinement (T-FDR) framework that formulates segmentation as a progressive refinement process. Specifically, the framework consists of three sequential stages: suppressing unreliable supervision, recovering boundary-sensitive features, and aligning latent semantic distributions. A Trust-guided Feature Structuring (TFS) module reduces noisy supervision by estimating pseudo-label reliability, followed by a Frequency-aware Residual Refinement (FRRF) module that enhances high-frequency boundary details in uncertain regions. Finally, a Bidirectional Diffusion Correction Module (BDCM) progressively refines latent semantic representations to improve consistency between labeled and unlabeled data. Experiments on three public cardiac MRI benchmarks demonstrate that T-FDR consistently outperforms state-of-the-art methods in both overlap and boundary metrics, indicating its effectiveness for robust

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semi-supervised segmentation under limited annotations.

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1. Introduction

Medical image segmentation plays a pivotal role in clinical diagnosis and treatment planning, and deep learning-based methods have achieved remarkable progress in recent years [1, 2]. In cardiac MRI, accurate segmentation is clinically important because it directly supports the quantitative assessment of ventricular volume, myocardial mass, stroke volume, and ejection fraction, which are key indicators for diagnosing cardiomyopathy, evaluating cardiac function, monitoring disease progression, and planning treatment. However, these approaches typically rely on large-scale annotated datasets, which are expensive to obtain in medical domains due to the need for expert labeling and privacy constraints [3]. In practice, pixel-wise annotation of a cardiac MRI study usually requires experienced cardiologists or radiologists to delineate the left ventricle, right ventricle, and myocardium across multiple end-diastolic and end-systolic slices, often taking tens of minutes for a single case. This high annotation cost substantially limits the availability of dense labels in routine clinical datasets. Semi-supervised learning (SSL) has therefore emerged as a promising solution to leverage both limited labeled data and abundant unlabeled data [4].

Despite these advances, semi-supervised segmentation remains particularly challenging in cardiac MRI. Compared with natural images, cardiac MRI often exhibits blurred anatomical boundaries, low tissue contrast, significant inter-slice structural variation, and distribution shifts across imaging centers and vendors [5, 6]. These characteristics make it difficult for SSL models to learn stable and reliable representations under limited supervision, and further amplify the impact of noisy pseudo-labels during training. More importantly, segmentation errors in boundary regions can propagate to clinically relevant measurements: even small contour deviations along the endocardial or epicardial borders may bias ventricular cavity and myocardial area estimation, which in turn affects derived indices such as end-diastolic volume, end-systolic volume, stroke volume, and ejection fraction. Inaccurate quantification of these measures may influence disease grading, assessment of systolic dysfunction, and treatment decisions such as follow-up planning or intervention timing. Such errors are particularly problematic in low-contrast regions where the delineation of the endocardium and epicardium is already ambiguous, further underscoring the need for boundary-sensitive and reliable segmentation under limited annotations.

Existing SSL methods for medical image segmentation are predominantly built upon consistency regularization and pseudo-labeling paradigms, such as Mean Teacher [7], UA-MT [8], Pseudo-Labeling [9], ICT [10], FixMatch [11], UDA [12], and CPS [13]. While these methods have demonstrated promising performance, they suffer from two fundamental limitations. First, pseudo-labels are typically treated uniformly without explicitly modeling their reliability, which leads to noisy supervision propagation and confirmation bias, especially in medical images with ambiguous boundaries and heterogeneous textures [14]. Second, most methods operate in the spatial domain and focus primarily on low-frequency structural information, while boundary-sensitive high-frequency details are insufficiently captured. Moreover, the latent semantic distribution of unlabeled data is often modeled in a static or coarse manner, limiting the ability to capture anatomical variability and resulting in suboptimal representation learning. Although recent works [15–17] have attempted to address these issues using consistency constraints, graph-based modeling, or prototype learning, they either lack adaptability to dynamic feature spaces or incur high computational complexity.

To address the above challenges, we argue that robust semi-supervised cardiac MRI segmentation requires jointly addressing three key aspects: pseudo-label reliability, boundary-sensitive feature recovery, and latent semantic distribution refinement. Based on this insight, we propose a trust-guided frequency-aware diffusion refinement (T-FDR) framework that organizes the learning process into three progressively connected stages.

First, a Trust-guided Feature Structuring (TFS) module is introduced to estimate pseudo-label reliability and suppress unreliable supervision signals, thereby reducing error accumulation and providing a stable foundation for representation learning. Second, a Frequency-aware Residual Refinement (FRRF) module operates on the structured representations to selectively enhance high-frequency components in uncertain regions, enabling improved boundary delineation while preserving structural consistency in confident areas. Finally, a Bidirectional Diffusion Correction Module (BDCM) is designed to progressively refine latent semantic distributions by modeling interactions between weakly and strongly augmented samples, allowing pseudo-labels to gradually approach the underlying true distribution and improving generalization performance.

The most closely related work to T-FDR is DiffRect [18], which also employs diffusion-based rectification to improve pseudo-label reliability. However, DiffRect primarily treats diffusion as a latent pseudo-label correction mechanism after pseudo labels have been generated, without explicitly modeling pseudo-label reliability, boundary-aware feature enhancement, or progressive semantic alignment during refinement. In particular, it lacks a trust-guided strategy to suppress unreliable pseudo-label regions before diffusion correction, and does not incorporate frequency-aware refinement to enhance high-frequency boundary information in uncertain areas. In contrast, T-FDR formulates pseudo-label refinement as a progressive process: trust-guided feature structuring first estimates pixel-wise reliability, frequency-aware residual refinement then enhances boundary-sensitive representations, and bidirectional diffusion correction finally aligns weakly augmented, strongly augmented, and labeled semantic distributions within a shared latent space. Thus, T-FDR extends diffusion-based rectification into a unified progressive refinement framework that jointly addresses reliability modeling, boundary-aware enhancement, and latent semantic alignment.

Specifically, our contributions are summarized as follows:

Trust-guided Feature Structuring (TFS): We introduce an uncertainty-aware structuring strategy that adaptively reweights unreliable pseudo-label regions and enforces semantic consistency in the latent feature space, improving robustness against noisy supervision. Frequency-aware Residual Refinement (FRRF): We propose a frequency-domain refinement mechanism that selectively enhances high-frequency boundary details in uncertain regions while preserving stable structures in confident areas. Bidirectional Diffusion Correction Module (BDCM): We design a diffusion-based latent alignment strategy that progressively refines pseudo-label distributions by modeling interactions between weak and strong augmentations, improving representation quality and robustness under the evaluated data distributions.

2. Related Work

2.1. Semi-supervised learning

Semi-supervised learning (SSL) aims to improve model generalization by leveraging abundant unlabeled data alongside a limited number of annotated samples, which is particularly important in medical image segmentation where expert annotations are costly to obtain [4]. Existing SSL methods are largely built upon two paradigms: consistency regularization and pseudo-label learning. Consistency-based approaches assume that model predictions should remain stable under different perturbations. Mean Teacher [7] enforces this by introducing an exponential moving average teacher model to guide student predictions, while FixMatch [11] combines consistency learning with confidence-based pseudo-label filtering to reduce noise from low-confidence regions. CPS [13] further extends this idea using dual networks that supervise each other through cross pseudo-labeling, improving training stability. More recent studies have explored uncertainty-guided teacher-student consistency for semi-supervised MRI segmentation [19] and uncertainty-based pseudo-label refinement for semi-supervised volumetric medical image segmentation [20]. Building on these frameworks, recent studies incorporate structural awareness and feature-level alignment to better handle complex anatomical variations. Multidimensional consistency learning [21] explicitly models structural relationships across multiple spatial views, while feature consistency-aware methods [22] enforce alignment in the feature space to reduce discrepancies between labeled and unlabeled samples. To address sensitivity to perturbation strength, differential perturbation consistency [23] introduces multi-level augmentations to extract more reliable supervision, and strong-teacher frameworks [24] enhance teacher robustness through stronger augmentations. In parallel, pseudo-label refinement has been extensively studied. Collaborative self-training [25] iteratively improves pseudo-label quality via multi-model cooperation, and mixup-based approaches such as MMISeg [26] smooth decision boundaries by interpolating between samples. More recently, transformer-based and foundation-model-assisted semi-supervised segmentation methods have shown promising performance in medical image analysis. CPC-SAM [27], for example, introduces cross-prompting consistency with the Segment Anything Model (SAM) to exploit promptable transformer representations for semi-supervised medical image segmentation. This approach is complementary to T-FDR: CPC-SAM focuses on consistency between complementary SAM prompts, whereas T-FDR focuses on trust-guided supervision, frequency-aware boundary refinement, and bidirectional latent diffusion correction. Despite these advances, several limitations remain. Most methods treat pseudo labels uniformly without explicitly modeling pixel-wise reliability, making them vulnerable to noise propagation and confirmation bias. Moreover, they primarily operate in the spatial domain and struggle to capture fine-grained boundary details. Finally, the latent semantic discrepancy between labeled and unlabeled data is rarely addressed, leading to suboptimal representation consistency, especially in challenging medical scenarios such as cardiac MRI.

2.2. Semi-Supervised Cardiac MRI Segmentation

Semi-supervised cardiac MRI segmentation introduces additional challenges due to the intrinsic properties of the data. Compared with natural images, cardiac MRI often exhibits low tissue contrast, ambiguous boundaries between adjacent structures, significant inter-slice anatomical variation, and domain shifts across acquisition centers and vendors [5]. These factors make pseudo labels more error-prone in uncertain regions and weaken the effectiveness of consistency constraints across views. To address these challenges, recent studies incorporate domain-specific priors into SSL frameworks. Multidimensional consistency learning [21] imposes structural constraints across multiple planes to improve stability under anatomical variations, while feature consistency-aware approaches [22] align semantic representations of cardiac structures across augmentations. Differential perturbation consistency [23] and strong-teacher strategies [24] further improve robustness by designing augmentation schemes tailored to cardiac MRI characteristics. Self-training methods have also been adapted to this task: collaborative self-training [25] reduces bias through multi-model refinement, and MMISeg [26] improves segmentation near low-contrast boundaries using mixup-based interpolation. Beyond semi-supervised settings, recent advances in cardiac image analysis also demonstrate the importance of hybrid architectures and domain adaptation. For example, P2TC [28] combines pyramid pooling with Transformer–CNN modeling for accurate 3D whole-heart segmentation, while latent-space adversarial domain adaptation [29] improves cardiac MRI segmentation under cross-domain distribution shifts. These studies indicate that both structural representation learning and distribution alignment are critical for robust cardiac segmentation. To better capture structural details, frequency-domain modeling has been introduced. By leveraging high-frequency components that encode edge information, methods such as weakly semi-supervised cardiac MRI segmentation with frequency-domain pseudo-label dynamic mixed supervision [30] enhance boundary delineation in cardiac structures. However, these approaches typically rely on fixed spectral decomposition and lack adaptive modulation for highly uncertain regions. Generative models provide another direction for modeling complex distributions. Early work adopts GAN-based frameworks [31], while recent studies explore diffusion models such as DiffRect [18] to refine pseudo labels in latent space. Nevertheless, existing generative approaches mainly operate at the image or mask level and do not explicitly align semantic representations across weakly augmented, strongly augmented, and labeled data in a unified latent space. As a result, current methods still suffer from three key limitations: insufficient modeling of pseudo-label reliability, lack of adaptive boundary-aware feature enhancement, and inadequate alignment of latent semantic distributions. These gaps motivate the proposed T-FDR framework, which integrates trust-guided supervision, frequency-aware refinement, and diffusion-based latent correction into a progressive and unified solution.

3. Methods

Our main model approach is semi-supervised learning (SSL), aiming to enhance the generalization ability of the model by fully leveraging a large amount of unlabeled data with only a small number of labeled samples. Our labeled dataset is: $\mathcal{D}_l = \{(\mathrm{I}_i, l_i)\}_{i=1}^{N_l}$, where $\mathrm{I}_i \in \mathbb{R}^{H\times W}$ represents the input medical image, $l_i \in \{0, 1, \ldots, C\}^{H\times W}$ represents the corresponding true segmentation label. Unlabeled datasets are denoted as $\mathcal{D}_u = \{\mathrm{I}_j\}_{j=1}^{N_u}$, satisfying $N_u \gg N_l$. The objective of SSL is to optimize a segmentation model $f_\theta(\cdot)$ parameterized by θ, by jointly exploiting $\mathcal{D}_l$ and $\mathcal{D}_u$. For unlabeled samples $\mathrm{I} \in \mathcal{D}_u$, pseudo labels are generated using the current model predictions: $\hat{y} = \arg\max f_\theta(\mathrm{I})$, where $\hat{y}$ denotes the hard pseudo label inferred from the model output. To mitigate the adverse impact of noisy pseudo labels, only predictions with high confidence are retained. Specifically, a confidence score $p_{\max}$ is computed as $p_{\max} = \max_c f_\theta^c(\mathrm{I})$. Consistency regularization enforces the model to produce invariant predictions under different perturbations. Given an unlabeled image I, two stochastic augmentations are applied by $\mathrm{I}_w = \mathcal{A}_w(\mathrm{I}), \quad \mathrm{I}_s = \mathcal{A}_s(\mathrm{I})$, where $\mathcal{A}_w(\cdot)$ and $\mathcal{A}_s(\cdot)$ represent weak and strong augmentations, respectively. The model predictions are o_w and o_s. A consistency loss is defined to minimize the discrepancy between the two predictions:

$$\mathcal{L}_{\text{cons}} = \mathbb{E}_{\mathrm{I}\sim\mathcal{D}_u} \left\| o_s - o_w \right\|_2^2 . \tag{1}$$

For labeled data, a supervised segmentation loss $\mathcal{L}_{\text{sup}}$ is applied:

$$\mathcal{L}_{\text{sup}} = \mathbb{E}_{(\mathrm{I},l)\sim\mathcal{D}_l} \ell \left[f_\theta(\mathrm{I}), l \right], \tag{2}$$

where $\mathbb{E}$ represents the expectation on the data distribution and stochastic data augmentations, approximated by averaging each mini-batch in practice.

The overall training objective integrates both supervised and unsupervised terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda(t)\mathcal{L}_{\text{cons}}, \tag{3}$$

where $\lambda(t)$ is a time-dependent weighting function that gradually increases as training progresses, ensuring stable optimization during early training stages.

3.1. Preliminaries: Diffusion Models

The diffusion model consists of a forward diffusion process and a reverse denoising process. The forward process is defined as a discrete Markov chain of length T, and for each step $t \in [1, T]$, the diffusion model adds noise ϵ_t randomly sampled from a standard Gaussian distribution $\mathcal{N}(0, 1)$ to gradually corrupt the input image I_0. Ultimately, a pure noise image I_T is generated, consisting entirely of Gaussian noise. Since the noisy image I_t at time step t only depends on the noisy image I_{t-1} from the previous time step, this Markov chain process can be expressed as:

$$q(\mathrm{I}_{1:T}|\mathrm{I}_0) = \prod_{t=1}^{T} q(\mathrm{I}_t|\mathrm{I}_{t-1}), \tag{4}$$

$$q(\mathrm{I}_t|\mathrm{I}_{t-1}) = \mathcal{N}\left(\mathrm{I}_t; \sqrt{1-\beta_t}\mathrm{I}_{t-1}, \beta_t\mathbf{I}\right), \tag{5}$$

where $q(\mathrm{I}_t|\mathrm{I}_{t-1})$ represents the forward diffusion process from the noise image I_{t-1} to the noise image I_t, and $\prod_{t=1}^{T} q(\mathrm{I}_t|\mathrm{I}_{t-1})$ denotes the progressive noise addition. $\beta_t \in (0, 1)$ represents the variance schedule across diffusion steps, and $\mathbf{I}$ denotes the identity matrix. Notably, instead of generating samples sequentially, we can obtain I_t at any time step in closed form as: $q(\mathrm{I}_t|\mathrm{I}_0) = \mathcal{N}\left(\mathrm{I}_t; \sqrt{\bar{\alpha}_t}\mathrm{I}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$, where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ and $\alpha_t = 1 - \beta_t$. Thus, I_t can be reparametrized as:

$$\mathrm{I}_t = \sqrt{\bar{\alpha}_t}\mathrm{I}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon. \tag{6}$$

When the number of diffusion steps T is sufficiently large, $\bar{\alpha}_t$ approaches zero, and I_t becomes nearly indistinguishable from Gaussian noise. To reconstruct the clean sample, the reverse process approximates the posterior $q(\mathrm{I}_{t-1}|\mathrm{I}_t)$. Therefore, a neural network $\mu_\theta(\mathrm{I}_t, t)$ is introduced to parameterize the reverse transition:

$$p(\mathrm{I}_0, \ldots, \mathrm{I}_{T-1} \mid \mathrm{I}_T) = \prod_{t=1}^{T} p(\mathrm{I}_{t-1} \mid \mathrm{I}_t), \tag{7}$$

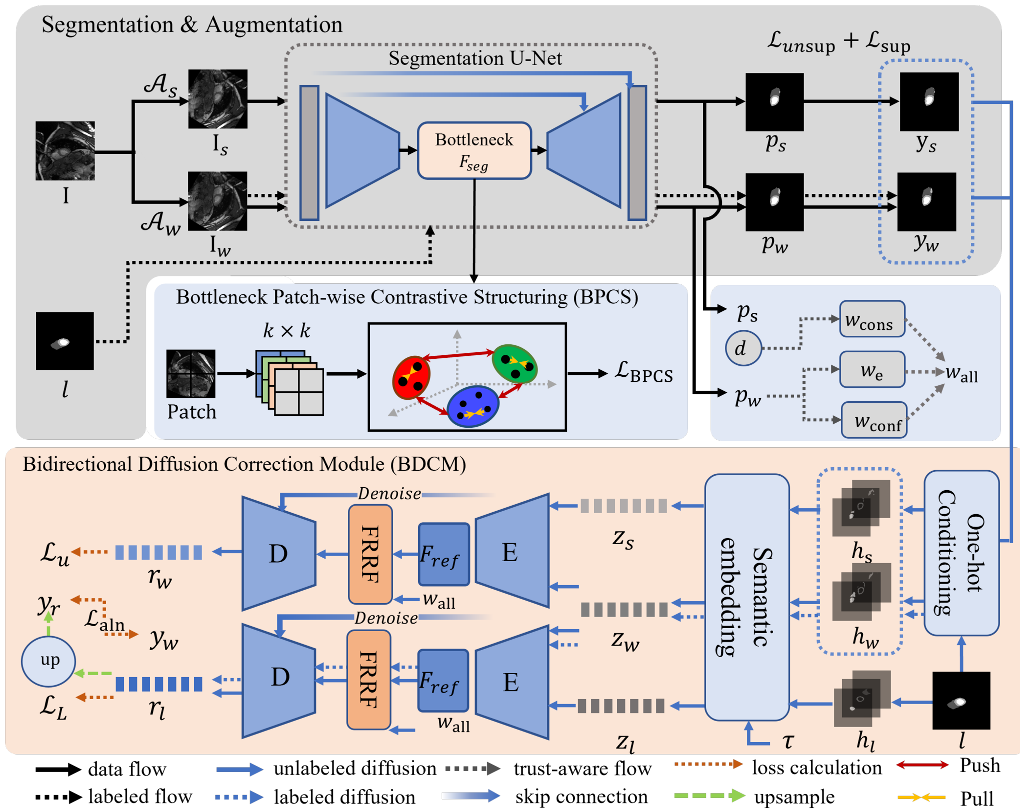

Figure 1: Overall framework of T-FDR.

$$p_\theta(\mathrm{I}_{t-1} \mid \mathrm{I}_t) = \mathcal{N}(\mathrm{I}_{t-1}; \mu_\theta(\mathrm{I}_t, t), \Sigma_\theta(\mathrm{I}_t, t)). \tag{8}$$

In this case, we adopt an x_0-prediction parameterization, where a prediction network $f_\theta(\mathrm{I}_t, t)$ directly estimates the clean sample from the noisy input:

$$\hat{\mathrm{I}}_0 = f_\theta(\mathrm{I}_t, t). \tag{9}$$

The reverse-process mean is then computed from the posterior $q(\mathrm{I}_{t-1} \mid \mathrm{I}_t, \hat{\mathrm{I}}_0)$:

$$\mu_\theta(\mathrm{I}_t, t) = \frac{\beta_t\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\hat{\mathrm{I}}_0 + \frac{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1-\bar{\alpha}_t}\mathrm{I}_t. \tag{10}$$

Following the variational inference framework, we optimize the clean-sample reconstruction network by minimizing the variational lower bound (LVB) on the negative log-likelihood.

We can train the model by optimizing the Evidence Lower Bound (ELBO) of the negative log-likelihood $-\log p_\theta(\mathrm{I}_0)$:

$$\mathbb{E}_{q(\mathrm{I}_0)}[-\log p_\theta(\mathrm{I}_0)] \leq \mathbb{E}_{q(\mathrm{I}_0,\mathrm{I}_{1:T})}\left[-\log \frac{p_\theta(\mathrm{I}_{0:T})}{q(\mathrm{I}_{1:T} \mid \mathrm{I}_0)}\right] = \mathcal{L}_{\mathrm{VLB}}. \tag{11}$$

Expanding the above inequality, a more specific form of the loss function can be obtained:

$$\mathcal{L}_{\mathrm{VLB}} = \mathbb{E}_q\left[L_T + \sum_{t>1} L_{t-1} - L_0\right], \tag{12}$$

$$L_T = D_{\mathrm{KL}}(q(\mathrm{I}_T \mid \mathrm{I}_0), p(\mathrm{I}_T)), \tag{13}$$

$$L_{t-1} = D_{\mathrm{KL}}(q(\mathrm{I}_{t-1} \mid \mathrm{I}_t, \mathrm{I}_0), p_\theta(\mathrm{I}_{t-1} \mid \mathrm{I}_t)), \tag{14}$$

$$L_0 = \log p_\theta(\mathrm{I}_0 \mid \mathrm{I}_1), \tag{15}$$

where D_{KL} represents the KL divergence between two Gaussian distributions. Ho et al. (2020) derived a more concise loss function in Denoising Diffusion Probabilistic Models. Under the x_0-prediction parameterization, the final objective function can be simplified to:

$$\mathcal{L}_{\mathrm{simple}}(\theta) = \mathbb{E}_{t,\mathrm{I}_0,\epsilon}\left\|\mathrm{I}_0 - f_\theta(\sqrt{\bar{\alpha}_t}\mathrm{I}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\right\|^2, \tag{16}$$

where t is uniformly sampled from $\{1, 2, \ldots, T\}$, $\mathrm{I}_0 \sim q(\mathrm{I}_0)$, $\epsilon \sim \mathcal{N}(0, 1)$.

3.2. Overview

The upper part of Fig. 1 is the Segmentation Network, while the bottom part corresponds to the Bidirectional Diffusion Correction Module. The symbols used in Fig. 1 are summarized in Table 1. Overall, T-FDR follows a progressively connected three-stage optimization strategy: TFS first suppresses unreliable supervision, FRRF then performs uncertainty-aware frequency-domain residual enhancement to restore boundary-sensitive details in uncertain regions, and BDCM finally corrects the remaining latent semantic discrepancy through explicit latent diffusion and feeds the rectified pseudo labels back to the segmentation network.

Table 1: Symbols in Fig. 1 and their descriptions

Symbol	Descriptions
$\mathrm{I}_w, \mathrm{I}_s$	weakly and strongly augmented input images
o_w, o_s	segmentation logits of the weak and strong branches
p_w, p_s	softmax probability maps of the weak and strong branches
y_w, y_s	hard pseudo labels obtained from confidence-thresholded predictions
h_w, h_s, h_l	one-hot semantic conditions constructed from weak pseudo labels, strong pseudo labels, and ground-truth labels
τ	conditioning scalar derived from Dice discrepancy and used to modulate latent correction
r_l^t	noisy label latent at diffusion timestep t
$\hat{r}_l^0$	predicted clean label latent reconstructed by the diffusion denoising network
z_w, z_s, z_l	semantic latent embeddings of weak, strong, and label branches
$w_e, w_{\text{cons}}, w_{\text{conf}}$	entropy-based, perturbation-consistency, and confidence-gate weights
$w_{\text{trust}}, w_{\text{bnd}}, w_{\text{all}}$	uncertainty-aware trust weight, boundary-emphasis weight, and final pixel-wise unsupervised weight
w_{ref}	rectification weight computed from refined prediction entropy, confidence, and boundary emphasis
y_r	hard rectified pseudo label derived from the refined probability map
$\mathcal{L}_{\text{sup}}, \mathcal{L}_{\text{unsup}}$	supervised and trust-weighted unsupervised segmentation losses
$\mathcal{L}_{\text{BPCS}}$	bottleneck patch-wise contrastive structuring loss
$\mathcal{L}_{\text{U}}, \mathcal{L}_{\text{L}}$	unlabeled and labeled latent reconstruction losses
$\mathcal{L}_{\text{ref}}^{\text{sup}}, \mathcal{L}_{\text{ref}}^{\text{unsup}}, \mathcal{L}_{\text{rect}}$	refinement supervision on labeled data, refinement supervision on unlabeled data, and delayed rectification loss

3.3. Trust-aware Feature Structuring

3.3.1. Trust-weighted Supervision

The pseudo label y_w, y_s are defined in 3.5 as the class with the maximum predicted probability. However, this hard assignment ignores predictive uncertainty. To quantify uncertainty, we compute the Shannon entropy:

$$U = -\sum_{c=1}^{C} p_w \log p_w, \tag{17}$$

where $p_\theta^c(\mathrm{I}_w)$ denotes the predicted probability of class c at pixel location $\mathrm{I}_w \in \Omega$, and Ω represents the spatial domain of the weakly augmented image.

Since entropy lies in the range $[0, \log C]$, we normalize it to the interval $[0, 1]$ to obtain a consistent $\tilde{U}$. The entropy-based trust weight is then defined as

$$w_e = (1 - \tilde{U})^\alpha, \tag{18}$$

where $\alpha > 0$ is a hyperparameter controlling the sharpness of the confidence mapping. To capture instability introduced by strong perturbations, we define a consistency-based weight:

$$w_{\text{cons}} = \exp(-\beta d), \tag{19}$$

where $d = \|p_w - p_s\|$ measures the discrepancy between weak and strong predictions, and β controls the sensitivity to their disagreement.

To further eliminate extremely unreliable pixels, we introduce a confidence floor:

$$w_{\text{conf}} = \begin{cases} 1, & \max_c p_w^c > \varphi, \\ 0, & \text{otherwise}, \end{cases} \tag{20}$$

where φ is a predefined confidence threshold.

The uncertainty-aware trust weight is first defined as:

$$w_{\text{trust}} = w_e \cdot w_{\text{cons}} \cdot w_{\text{conf}}. \tag{21}$$

To further emphasize structurally ambiguous contour regions, we introduce a boundary-aware factor w_{bnd}, computed from the morphological gradient of the pseudo label. The final pixel-wise weight becomes:

$$w_{\text{all}} = w_{\text{trust}} \cdot w_{\text{bnd}}. \tag{22}$$

The reweighted unsupervised objective becomes:

$$\mathcal{L}_{\text{unsup}} = \mathbb{E}_u \left[w_{\text{all}} \cdot \ell(p_u, y) \right], \tag{23}$$

where $\ell(\cdot, \cdot)$ denotes a combination of Cross-Entropy and Dice loss, y denotes the pseudo label, and p_u denotes the unlabeled predictions.

Relation to uncertainty estimation.. The proposed weighting is closely related to established uncertainty estimation strategies in semi-supervised learning. The entropy term $w_e = (1 - \tilde{U})^{\alpha}$ is a monotone decreasing transform of predictive entropy, so pixels with flatter posterior distributions receive smaller weights. This is consistent with the common use of entropy as a surrogate for prediction uncertainty in pseudo-label filtering and confidence-aware learning. The consistency term $w_{\text{cons}} = \exp(-\beta d)$ measures the agreement between weakly and strongly perturbed predictions, and is therefore related to perturbation-based uncertainty estimation, where instability across views indicates reduced local reliability. The confidence gate w_{conf} further excludes extremely low-confidence predictions from participating in pseudo supervision.

Accordingly, w_{trust} can be interpreted as a reliability-aware reweighting factor on unlabeled pixels. When pseudo-label errors are more likely to occur in regions with high entropy or large weak–strong discrepancy, this reweighting suppresses the contribution of unreliable pseudo labels to the optimization process. The additional factor w_{bnd} does not estimate uncertainty itself; instead, it redistributes supervision toward contour regions, where cardiac MRI segmentation is both clinically important and more error-prone. Therefore, the final weight w_{all} combines uncertainty-aware reliability control with boundary-aware emphasis, which is consistent with the actual training objective used in our implementation.

3.3.2. Bottleneck Patch-wise Contrastive Structuring

Given an input image I, we extract the bottleneck feature representation $F_{seg} = \varepsilon_{seg}(\mathrm{I})$, where $F_{seg} \in \mathbb{R}^{B\times C\times H'\times W'}$. Here, C denotes the channel dimension and H', W' the spatial resolution of the bottleneck layer.

We divide the spatial dimensions of F_{seg} into a grid of non-overlapping patches of size $k \times k$. For each patch feature F_i, we apply average pooling followed by a small Multi-Layer Perceptron projection head $\psi(\cdot)$ to obtain a normalized embedding vector:

$$e_i = \psi(\text{AvgPool}(F_i)), \quad e_i \in \mathbb{R}^d, \tag{24}$$

where d is the dimension of the contrastive embedding. Each embedding e_i is associated with the pseudo label y of its corresponding patch.

For a given anchor embedding e_i, we define its positive set as all other embeddings originating from patches predicted to belong to the same class:

$$\mathcal{P}(i) = \{k \mid y^k = y^i, k \neq i\}, \tag{25}$$

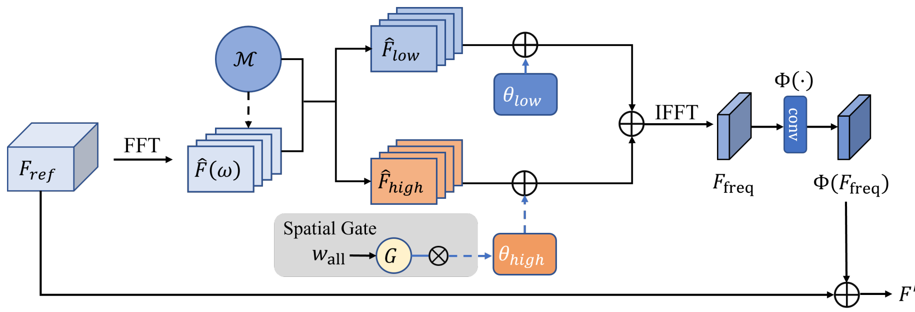

Figure 2: Frequency-aware Residual Refinement Framework (FRRF)

where the symbols y^i and y^k denote the patch-level class labels for the i-th and k-th image patches. The supervised contrastive loss for this anchor is then defined as:

$$\mathcal{L}_{\text{con}} = -\frac{1}{|\mathcal{P}(i)|} \sum_{k \in \mathcal{P}(i)} \log \frac{\exp(e_i^T e_k / \eta)}{\sum_j \exp(e_i^T e_j / \eta)}, \tag{26}$$

where $\eta > 0$ is a temperature scaling parameter, and $e_i^T e_k$ denotes the cosine similarity between the L2-normalized embeddings e_i and e_k. The denominator sum is typically computed over a large batch of embeddings.

The final Bottleneck Patch-wise Contrastive Structuring loss is computed separately for weak and strong augmentations and averaged:

$$\mathcal{L}_{\text{BPCS}} = \frac{1}{2} \left(\mathcal{L}_{\text{con}}^{w} + \mathcal{L}_{\text{con}}^{s} \right). \tag{27}$$

The BPCS loss is integrated into the overall segmentation objective with a weighting factor λ_B:

$$\mathcal{L}_{seg} = \mathcal{L}_{\text{sup}} + \lambda_u \mathcal{L}_{\text{unsup}} + \lambda_B \mathcal{L}_{\text{BPCS}}. \tag{28}$$

3.4. Frequency-Aware Residual Refinement Framework

In this section, we propose a Frequency-aware Residual Refinement Framework (FRRF) that operates directly in the latent semantic space and integrates uncertainty-aware modulation with frequency-domain residual refinement [32, 33]. FRRF itself does not introduce an explicit forward or reverse diffusion process; instead, it enhances boundary-sensitive details and suppresses unreliable high-frequency responses before the downstream latent diffusion correction performed by BDCM.

Let the refinement network produce a bottleneck feature representation $F_{\text{ref}} \in \mathbb{R}^{B\times C\times H'\times W'}$, where C denotes the channel dimension, and H', W' denote the spatial resolution of the bottleneck layer.

We transform the latent feature F_{ref} into the frequency domain using the standard 2D discrete Fourier transform (DFT):

$$\hat{F}(u,v) = \mathcal{F}(F_{\text{ref}}) = \sum_{h=0}^{H'-1}\sum_{w=0}^{W'-1} F_{\text{ref}}[h,w]\cdot \exp(-j2\pi\left(\frac{uh}{H'} + \frac{vw}{W'}\right)), \tag{29}$$

where $\mathcal{F}(\cdot)$ denotes the Fourier transform operator.

To achieve explicit decoupling of frequency components, we introduce a simple radial frequency mask $\mathcal{M}(u,v)$. This mask is centered at the frequency domain origin and performs a binary partition based on the frequency radius:

$$\mathcal{M}(u,v) = \begin{cases} 1, & \text{if } \sqrt{u^2+v^2} \le r_c, \\ 0, & \text{otherwise} \end{cases} \tag{30}$$

Here, r_c is a pre-defined cutoff frequency radius that delineates the boundary between low and high frequencies. Thus, we can decompose $\hat{F}$ into a low-frequency component $\hat{F}_{low}$ and a high-frequency component $\hat{F}_{high}$:

$$\begin{cases} \hat{F}_{low} = \mathcal{M}(u,v)\odot \hat{F}(u,v), \\ \hat{F}_{high} = (1-\mathcal{M}(u,v))\odot \hat{F}(u,v). \end{cases} \tag{31}$$

Thus, the original frequency representation can be reconstructed as the sum of the two parts: $\hat{F} = \hat{F}_{low} + \hat{F}_{high}$.

We introduce learnable scalar parameters ϕ_{low} and ϕ_{high} for the decomposed low and high-frequency components, from which the effective modulation coefficients are defined as:

$$\begin{cases} \theta_{low} = 1 + \delta\tanh(\phi_{low}), \\ \theta_{high}^{\text{base}} = 1 + \delta\tanh(\phi_{high}), \end{cases} \tag{32}$$

where $\phi_{low}, \phi_{high} \in \mathbb{R}$ are learnable scalar parameters, θ_{low} and $\theta_{high}^{\text{base}}$ denote the corresponding effective modulation coefficients, $\tanh(\cdot)$ confines the output to the interval $(-1,1)$, and δ is a positive hyperparameter controlling the maximum enhancement or attenuation strength.

To incorporate spatial uncertainty, we leverage the trust weight w_{all} defined in 3.3 and define a spatial frequency gate G that modulates the high-frequency gain:

$$G = 1 - w_{\text{all}}, \quad \theta_{high} = 1 + G \cdot (\theta_{high}^{\text{base}} - 1), \tag{33}$$

Given the decomposed low and high frequency components in the Fourier domain, the modulated spatial representation is reconstructed as:

$$F_{\text{freq}} = \mathcal{F}^{-1}\left(\theta_{low}\hat{F}_{low} + \theta_{high}\hat{F}_{high}\right), \tag{34}$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the inverse Fourier transform.

To preserve semantic consistency while enabling targeted correction in uncertain regions, we adopt a residual refinement strategy:

$$F' = F_{\text{ref}} + \Phi(F_{\text{freq}}), \tag{35}$$

where $\Phi(\cdot)$ denotes a lightweight feed-forward transformation.

This residual formulation is a deterministic frequency-domain refinement step rather than a diffusion process. Its role is to deliver uncertainty-aware, boundary-enhanced latent features to the subsequent BDCM, where explicit latent diffusion correction is performed.

Why frequency-aware refinement enhances boundaries.. The effectiveness of FRRF can be interpreted directly from the implemented operator. In cardiac MRI, ventricular and myocardial boundaries correspond to rapid spatial intensity transitions, whereas homogeneous cavities and myocardium are dominated by slower spatial variation. After mapping the bottleneck feature to the Fourier domain, these two types of information are separated by a radial mask into low- and high-frequency bands. Consequently, modifying the high-frequency band primarily affects contour-sensitive responses, while modifying the low-frequency band primarily affects coarse anatomical layout.

The implemented module applies bounded per-channel gains to both bands, with gain values constrained by a tanh parameterization. More importantly, the spatial gate derived from the trust map modulates only the deviation of the high-frequency gain from unity. Therefore, uncertain spatial locations receive stronger adaptive changes in high-frequency amplitude, whereas reliable locations remain closer to the ungated response. Since boundary ambiguity in cardiac MRI is concentrated near endocardial and epicardial interfaces, this design increases the sensitivity of the representation to boundary-relevant spectral components specifically where delineation is difficult, rather than uniformly amplifying all spatial regions.

The residual structure further limits the effect of this operation. After inverse Fourier transform, the modulated feature is passed through a lightweight feed-forward mapping and then added back to the original bottleneck feature. This means that FRRF acts as a bounded residual correction rather than a full replacement of the latent representation. In combination with the bounded spectral gains, this design constrains the magnitude of the refinement while preserving the coarse semantic structure already encoded in the bottleneck representation. Hence, the practical effect of FRRF is an uncertainty-localized enhancement of contour-sensitive components with controlled disturbance to region-level semantics.

3.5. Bidirectional Diffusion Correction Module

Although TFS and FRRF mitigate unreliable supervision and enhance boundary details, pseudo labels from weak and strong views may still contain latent semantic discrepancies. To address this, we propose a Bidirectional Diffusion Correction Module (BDCM) that performs diffusion-based correction in the semantic latent space—not at the image level—progressively transporting noisy pseudo-label semantics toward more reliable targets.

Semantic-aligned one-hot conditioning. After obtaining the predictions o_w and o_s, we first generate pseudo labels y_w and y_s by confidence-thresholded argmax. To construct a unified semantic condition across weak, strong, and labeled branches, we convert these discrete labels into one-hot maps:

$$h_w = \mathbb{I}(y_w = c), \quad h_s = \mathbb{I}(y_s = c), \quad h_l = \mathbb{I}(l = c), \tag{36}$$

where $h_w, h_s, h_l \in \mathbb{R}^{C\times H\times W}$ denote the weak pseudo-label condition, the strong pseudo-label condition, and the ground-truth condition, respectively.

To obtain semantic latent representations, we introduce a semantic embedding module $E_{\text{sem}}(\cdot)$, which maps the one-hot semantic masks into the latent space:

$$z_w = E_{\text{sem}}(h_w), \quad z_s = E_{\text{sem}}(h_s), \quad z_l = E_{\text{sem}}(h_l), \tag{37}$$

where $z_w, z_s, z_l \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{16}\times D}$, and D denotes the latent channel dimension. In practice, the diffusion timestep t is embedded by sinusoidal positional encoding and jointly injected into the denoising network together with the corresponding semantic latent condition. This design ensures that semantic information from weak, strong, and labeled branches is represented in a consistent latent space before diffusion correction.

Overview of bidirectional correction. BDCM consists of two complementary latent diffusion paths: a **strong-to-weak** path for unlabeled data, which reconstructs the weak-view latent conditioned on the strong-view semantics, and a **weak-to-label** path for labeled data, which reconstructs the ground-truth latent conditioned on the weak-view semantics. Together, they progressively close the semantic gap among strong-view pseudo labels, weak-view pseudo labels, and reliable label semantics.

3.5.1. Strong-to-Weak diffusion for unlabeled data

For an unlabeled image, we extract strong- and weak-view semantic latent embeddings z_s and z_w. The strong-to-weak path corrects the weak latent representation under the guidance of z_s, explicitly modeling the semantic discrepancy between two perturbation views in latent space. A forward diffusion process is applied to the weak latent z_w:

$$z_w^t = \sqrt{\bar{\alpha}_t}\, z_w + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \tag{38}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, $\alpha_t = 1 - \beta_t$, and $\{\beta_t\}_{t=1}^{T}$ denotes a predefined noise schedule.

A conditional denoising network f_θ is then trained to reconstruct the clean weak latent from its noisy version, conditioned on the strong semantic latent. Under the adopted x_0-prediction parameterization, the network directly outputs the estimated clean latent

$$\hat{z}_w = f_\theta(z_w^t, t, z_s), \tag{39}$$

and the reverse transition is written as

$$p_\theta(z_w^{t-1} \mid z_w^t, z_s) = q\big(z_w^{t-1} \mid z_w^t, \hat{z}_w\big)\,. \tag{40}$$

The corresponding unlabeled latent reconstruction objective is:

$$\mathcal{L}_{\mathrm{U}} = \mathbb{E}_{z_w,t}\left[\|z_w - \hat{z}_w\|_2^2\right]. \tag{41}$$

This objective trains the diffusion model to learn a conditional semantic transport from the strong-view to the weak-view semantic distribution, improving cross-view semantic consistency for unlabeled data.

3.5.2. Weak-to-Label diffusion for labeled data

For labeled samples, the weak-view pseudo-label representation is still imperfect and may deviate from the ground-truth semantic structure. To bridge this gap, we introduce a weak-to-label diffusion path, where the weak latent z_w is used as the condition and the label latent z_l is used as the reconstruction target.

Specifically, we perturb the label latent by the same forward diffusion process:

$$z_l^t = \sqrt{\bar{\alpha}_t}\, z_l + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \tag{42}$$

The reverse denoising process is conditioned on the weak semantic latent. Under the same x_0-prediction parameterization, the denoising network directly estimates the clean label latent:

$$\hat{z}_l = f_\theta(z_l^t, t, z_w). \tag{43}$$

The reverse transition is then written as

$$p_\theta(z_l^{t-1} \mid z_l^t, z_w) = q\left(z_l^{t-1} \mid z_l^t, \hat{z}_l\right). \tag{44}$$

The corresponding labeled latent reconstruction loss is defined as:

$$\mathcal{L}_{\mathrm{L}} = \mathbb{E}_{z_l, t}\left[\|z_l - \hat{z}_l\|_2^2\right]. \tag{45}$$

This path drives weak pseudo-label semantics toward the reliable semantic center defined by ground-truth labels, reducing the gap between pseudo and real annotations.

3.5.3. Rectified pseudo-label generation

After training, the learned weak-to-label diffusion path is further used to rectify weak pseudo labels. Starting from Gaussian noise

$$r_l^T \sim \mathcal{N}(0, \mathbf{I}), \tag{46}$$

we perform progressive reverse diffusion conditioned on z_w. At each step, the denoising network first predicts the clean latent:

$$\hat{r}_l^0 = f_\theta(r_l^t, t, z_w), \tag{47}$$

and then computes the reverse update through the posterior mean induced by the predicted clean latent:

$$r_l^{t-1} = \frac{\beta_t \sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t} \hat{r}_l^0 + \frac{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1-\bar{\alpha}_t} r_l^t + \sigma_t \eta, \quad \eta \sim \mathcal{N}(0, \mathbf{I}). \tag{48}$$

In implementation, the recovered latent is decoded by the refinement network into segmentation logits, followed by a softmax probability map:

$$o_r = \mathrm{Dec}(\hat{r}_l^0), \qquad p_r = \mathrm{softmax}(o_r). \tag{49}$$

The final rectified pseudo label is then instantiated as a hard label by confidence thresholding and argmax selection:

$$y_r = \arg\max \left(p_r \odot \mathbb{I}(\max_c p_r^c > \varphi) \right). \tag{50}$$

Therefore, p_r denotes the refined soft probability map, whereas y_r denotes the hard rectified pseudo label that is subsequently used for delayed pseudo supervision of the segmentation network.

3.5.4. Coupling with trust-guided and frequency-aware refinement

BDCM serves as the final semantic correction stage within the progressive refinement framework. TFS first suppresses unreliable supervision via trust weight estimation, and FRRF then enhances boundary-sensitive high-frequency details in uncertain regions. Building on these, BDCM rectifies the residual latent semantic discrepancy across weak, strong, and labeled predictions, feeding the rectified pseudo label y_r back to the segmentation branch to form a closed-loop optimization spanning trust-guided supervision, boundary-aware refinement, and latent semantic correction.

Diffusion Settings. The proposed BDCM is implemented using a latent diffusion formulation defined on semantic latent representations. During training, the diffusion timestep is uniformly sampled as $t \in [1, T]$, where we set $T = 1000$. A linear noise schedule $\{\beta_t\}_{t=1}^T$ is adopted, with β_t increasing from 10^{-4} to 0.02. Accordingly, $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$.

The denoising network f_θ is shared across both diffusion paths, while different conditioning signals are used for the strong-to-weak and weak-to-label corrections. The network follows an x_0-prediction parameterization and directly estimates the clean latent representation from the noisy latent input. All latent representations are generated by the semantic embedding module E_{sem}, ensuring that diffusion operates in a unified latent space.

During inference, we employ deterministic DDIM sampling to progressively recover the rectified latent representation.

The diffusion module is jointly trained with the segmentation network. At each iteration, pseudo labels and trust maps produced by the segmentation branch are first converted into semantic latent representations via E_{sem}, which serve as inputs to BDCM. The diffusion correction is activated after a warm-up stage of 1000 iterations to ensure stable early training.

3.6. Loss Function

The training procedure consists of three sequential optimization steps in each iteration: (i) segmentation network update, (ii) refinement network update, and (iii) delayed rectification feedback to the segmentation network. Since these steps optimize different parameter groups with different supervision sources, the full training process is more accurately described stage by stage than by a single monolithic objective.

Stage I: segmentation network update. For labeled samples, the weak branch is optimized by the supervised segmentation loss

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{CE}}(o_w^l, l) + \mathcal{L}_{\text{Dice}}(p_w^l, l), \tag{51}$$

where the superscript l denotes labeled data.

For unlabeled samples, weak predictions are converted into hard pseudo labels y_w^u by confidence thresholding and argmax. The uncertainty-aware trust weight is computed as

$$w_{\text{trust}} = w_e \cdot w_{\text{cons}} \cdot w_{\text{conf}}, \tag{52}$$

and is further combined with the boundary-emphasis factor w_{bnd} to obtain

$$w_{\text{all}} = w_{\text{trust}} \cdot w_{\text{bnd}}. \tag{53}$$

The weighted unsupervised loss on unlabeled samples is defined as

$$\mathcal{L}_{\text{unsup}} = \mathcal{L}_{\text{wCE}}(o_s^u, y_w^u; w_{\text{all}}) + \mathcal{L}_{\text{wDice}}(p_s^u, y_w^u; w_{\text{all}}) + \lambda_{\text{comp}}\mathcal{L}_{\text{comp}}, \tag{54}$$

where the superscript u denotes unlabeled data, and $\mathcal{L}_{\text{comp}}$ is the complementary consistency term between weak and strong predictions.

To further structure bottleneck semantics, we introduce the bottleneck patch-wise contrastive structuring loss $\mathcal{L}_{\text{BPCS}}$, which is applied to patch embeddings from labeled and unlabeled samples, using ground-truth labels for labeled data and pseudo labels for unlabeled data. The Stage-I segmentation objective is therefore

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{sup}} + \lambda_u\mathcal{L}_{\text{unsup}} + \lambda_B\mathcal{L}_{\text{BPCS}}, \tag{55}$$

where λ_u is a ramped unsupervised weight and λ_B controls the contribution of bottleneck patch-wise contrastive structuring.

Stage II: refinement network update. The refinement model is trained through a labeled correction path and an unlabeled correction path.

For labeled samples, the weak pseudo-label condition is corrected toward ground-truth semantic supervision. This yields a labeled refinement loss

$$\mathcal{L}_{\text{ref}}^{\text{sup}} = \mathcal{L}_{\text{CE}}(o_r^l, l) + \mathcal{L}_{\text{Dice}}(p_r^l, l) + \mathcal{L}_{\text{L}}, \tag{56}$$

where $\mathcal{L}_{\text{L}}$ is the labeled latent reconstruction loss defined in Eq. (45).

For unlabeled samples, the strong pseudo-label condition is corrected toward the refined weak pseudo target. The unlabeled refinement loss is defined as

$$\mathcal{L}_{\text{ref}}^{\text{unsup}} = \mathcal{L}_{\text{wCE}}(o_{r,s}^u, y_{r,w}^u; w_{\text{all}}) + \mathcal{L}_{\text{wDice}}(p_{r,s}^u, y_{r,w}^u; w_{\text{all}}) + \lambda_{\text{comp}}^r \mathcal{L}_{\text{comp}}^r + \mathcal{L}_{\text{U}}, \tag{57}$$

where $y_{r,w}^u$ denotes the refined weak pseudo target for unlabeled data, and $\mathcal{L}_{\text{U}}$ is the unlabeled latent reconstruction loss defined in Eq. (41).

The Stage-II refinement objective is

$$\mathcal{L}_{\text{ref}} = \mathcal{L}_{\text{ref}}^{\text{sup}} + \lambda_r \mathcal{L}_{\text{ref}}^{\text{unsup}}, \tag{58}$$

where λ_r controls the unsupervised refinement contribution.

Stage III: delayed rectification feedback. After a warm-up stage of N_{warm} iterations, the refinement model is used in inference mode to generate refined probability maps p_r and the corresponding hard rectified pseudo labels y_r for unlabeled samples. Based on p_r, a rectification weight w_{ref} is computed from refined prediction entropy, confidence thresholding, and boundary emphasis. The delayed rectification loss is then defined as

$$\mathcal{L}_{\text{rect}} = \mathcal{L}_{\text{wCE}}(o_w^u, y_r^u; w_{\text{ref}}) + \mathcal{L}_{\text{wDice}}(p_w^u, y_r^u; w_{\text{ref}}), \tag{59}$$

which is applied only to unlabeled samples and is activated only when the training iteration exceeds N_{warm}.

Training schedule. Within each iteration, the segmentation network is first updated using $\mathcal{L}_{\text{seg}}$, the refinement network is then updated using $\mathcal{L}_{\text{ref}}$, and finally the segmentation network receives additional rectification supervision from $\mathcal{L}_{\text{rect}}$ after warm-up. In our implementation, $N_{\text{warm}} = 1000$ iterations.

Optimization and convergence discussion.. The proposed framework is optimized by sequential stochastic gradient updates to the segmentation model and the refinement model within each training iteration. The segmentation model is first updated using supervised loss, trust-weighted unsupervised loss, and contrastive regularization. The refinement model is then updated using latent reconstruction and trust-weighted rectification objectives. After a warm-up period, the rectified pseudo labels are fed back to further supervise the segmentation model. Therefore, the overall procedure is more accurately viewed as block-wise stochastic optimization of coupled non-convex objectives rather than optimization of a single closed-form loss in one step.

Even under this coupled training scheme, several implementation choices improve stability in a manner consistent with standard stochastic optimization analysis. First, the uncertainty-derived trust weights used in the unsupervised terms are detached from the gradient graph and bounded by construction, while the boundary emphasis term only rescales supervision spatially without introducing additional trainable feedback. Together, these weights limit the influence of unreliable pseudo labels while allocating greater emphasis to contour regions. Second, the frequency-aware refinement module applies bounded spectral gains through a tanh parameterization and injects the refined response through a residual pathway, which constrains the magnitude of feature perturbation passed to the decoder. Third, the rectification feedback is activated only after a warm-up stage, thereby reducing harmful early coupling between immature pseudo labels and the refinement branch.

Accordingly, while the present framework does not admit a strict global convergence guarantee, its update dynamics are consistent with the standard expectation that block-wise stochastic gradient methods approach a first-order stationary regime under bounded gradients, locally smooth objectives, and decaying step sizes. In this work, the role of the theoretical analysis is therefore to justify why the proposed training process is well-conditioned in practice: uncertainty-aware trust weighting suppresses unreliable supervision, boundary-aware emphasis redistributes learning toward structurally critical regions, bounded spectral modulation prevents excessive feature distortion, and delayed rectification avoids unstable early feedback loops.

4. Experiments

4.1. Datasets

To validate the performance of our T-FDR in cardiac segmentation task, we conducted experiments on the Automatic Cardiac Diagnosis Challenge (ACDC) dataset, a clinically annotated benchmark for cardiac MRI segmentation and pathology classification. In our implementation, we use 100 annotated subjects, uniformly distributed across five clinically defined subgroups: normal subjects (NOR), myocardial infarction with heart failure (MINF), dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and right ventricular abnormalities (RV). Each subject includes expert-annotated ground truth at the end-diastolic (ED) and end-systolic (ES) phases, yielding 200 frame-level cases in total and covering left/right ventricular endocardium, epicardium, and myocardium structures.

To further test the reliability of our model on the cardiac dataset, we extended our experiments to the Multi-sequence Cardiac MR Segmentation Challenge (MScmrseg19) dataset. This dataset contains multi-institutional, multi-sequence cardiac MR images acquired from patients with hypertrophic and dilated cardiomyopathy. It includes 45 subjects, each providing three MR sequences (LGE, T2, and bSSFP) with corresponding manual annotations for left ventricle (LV), myocardium (MYO), and right ventricle (RV). The dataset introduces significant domain shifts across sequences and scanners.

Additionally, we evaluated the robustness of our model to multi-vendor heterogeneity on the Multi-Centre, Multi-Vendor & Multi-Disease Cardiac Image Segmentation (M&MS) dataset. The M&MS dataset is designed to assess performance under inter-vendor and inter-center variability. It contains cardiac cine-MRI scans acquired from multiple clinical centers using scanners from four different vendors (Vendor A–D). The dataset comprises annotated cases covering diverse cardiac conditions, including normal subjects as well as patients with various cardiomyopathies. Each subject provides short-axis cine MRI sequences with expert manual annotations for three primary cardiac structures: left ventricle (LV), myocardium (MYO), and right ventricle (RV), typically at end-diastolic (ED) and end-systolic (ES) phases. Because all evaluated vendors are represented in the mixed development/evaluation protocol, this experiment is not a held-out-vendor or held-out-center domain-generalization study.

4.2. Experimental Setup

For fairness, all approaches are implemented under identical configurations and trained on an NVIDIA RTX 3090 GPU. On ACDC, we split the 100 annotated subjects at the subject level into 70, 10, and 20 subjects for training, validation, and testing, corresponding to 140, 20, and 40 frame-level ED/ES cases, respectively. We evaluate semi-supervised performance under 5% and 10% labeled protocols on this training split. For MScmrseg19, experiments are conducted with a 20% labeling ratio. On the M&MS dataset, we adopt a 20% labeled protocol to assess the stability of our framework under multi-vendor domain variations. All experiments are repeated with three random seeds, and we report the mean and standard deviation. To ensure fair comparison, all competing methods were evaluated under matched data splits, annotation ratios, training budgets, and evaluation protocols. Whenever official implementations or complete training settings were available, we followed the architectural configurations and optimization hyperparameters reported in the corresponding papers. For methods requiring dataset-specific adaptation or lacking complete implementation details, the hyperparameters were tuned on the validation set under the same protocol as our method, so that the reported comparison results were not obtained from mismatched or under-tuned settings.

4.3. Data Splits and Preprocessing

The data partitioning protocol was defined separately for each dataset before any 2D slice extraction, so that images and slices from one subject could not occur in more than one partition. For ACDC, the 100 annotated subjects were split at the subject level into 70 subjects for training, 10 for validation, and 20 for testing. The ED and ES phases were then selected within each partition, corresponding to 140, 20, and 40 frame-level ED/ES cases, respectively. The ACDC training set contains 1,312 2D slices in total. For MScmrseg19, we used the subject-level train/validation partitions provided by the challenge protocol and kept the two prescribed random partitions separate; the three sequences (LGE, T2, and bSSFP) were processed according to the dataset protocol, without applying the ACDC-specific ED/ES case definition. Results were averaged over the two partitions, and no subject or slice was moved between the corresponding training and validation sets. For M&MS, we used the subject-level development/evaluation partition supplied with the dataset and kept the evaluation subjects completely held out from training; its cine-MRI phases were processed according to the dataset protocol rather than being treated as ACDC-style ED/ES cases. Within the training subjects of each dataset, the labeled protocol was applied by selecting the specified fraction of training cases (5% or 10% for ACDC and 20% for MScmrseg19 and M&MS) as labeled data; all remaining training cases were used only as unlabeled data. The labeled subset was selected after the subject-level training partition had been fixed, and validation and test subjects were never used for supervised or unsupervised training. Thus, the reported results are based on fixed subject-level partitions and dataset-specific phase or sequence protocols rather than on slice-level random splits, and subject identifiers are not required to reproduce the protocol.

All images were converted to 2D short-axis slices and resized to 256×256 before being fed into the network. For images with intensity values larger than one, min–max normalization was applied to map intensities into $[0, 1]$. During training, we applied random rotation, random flipping, and weak–strong augmentation to construct paired views for semi-supervised consistency learning. The weak and strong branches share the same underlying patient-level split, while the strong branch additionally includes stronger perturbations such as intensity enhancement and Gaussian blur when enabled. During validation and testing, no random augmentation was applied; instead, the predicted masks were resized back to the original in-plane resolution before metric computation. This ensures that all reported Dice, Jac, HD95, and ASD values are computed in the original evaluation space.

4.4. Implementation Details

The hyperparameters are fixed to $\beta = 0.01$ and $\gamma = 0.80$. We adopt stochastic gradient descent (SGD) as the optimizer, with a momentum of 0.9, a weight decay of 0.0001, and an initial learning rate of 0.01. The batch size is set to 4, consisting of 2 labeled samples and 2 unlabeled samples. For all datasets, the training process is carried out for a total of 30,000 iterations. Our approach is compared against several state-of-the-art (SOTA) methods, including semi-supervised learning approaches such as UA-MT [8], URPC [34], MCNetV2 [35], FixMatch [11], SS-Net [36], AD-MT [37], and INCL [38], the recent transformer- and SAM-based semi-supervised method CPC-SAM [27], as well as the fully supervised baseline nnUNetv2. In each experimental setting, only a limited portion of data from a single domain is annotated, while the remaining data are treated as unlabeled samples. To quantitatively evaluate the segmentation performance, we employ four widely used metrics: Dice coefficient (Dice), Jac coefficient (Jac), 95% Hausdorff Distance (HD95), and Average Surface Distance (ASD).

Implementation details of T-FDR.. The segmentation backbone is a five-stage U-Net variant with encoder–decoder channel dimensions of $[16, 32, 64, 128, 256]$, skip connections at each resolution level, and bilinear upsampling disabled. Each convolutional block contains two 3×3 convolutions with padding one, BatchNorm2d, LeakyReLU, and dropout; the encoder dropout rates are $[0.05, 0.10, 0.20, 0.30, 0.50]$. The decoder contains four upsampling blocks with transposed-convolution upsampling, skip-feature concatenation, and the same convolutional-block pattern, followed by a 1×1 output convolution. No self-attention or cross-attention layer is used. The diffusion denoising branch is implemented as a latent-space U-shaped encoder–decoder with a sinusoidal timestep embedding of dimension 128 followed by two fully connected layers of dimensions $128 \rightarrow 512 \rightarrow 512$. For a 256×256 input, diffusion is applied to the bottleneck semantic latent tensor with 256 channels at the deepest encoder resolution (i.e., $256 \times 16 \times 16$ before any task-dependent spatial changes), rather than to the raw image. The diffusion schedule uses 1000 forward timesteps and 10 DDIM sampling steps. The bottleneck adaptor maps the concatenated encoder and denoised latent features from 512 to 256 channels. The frequency-band gate at the bottleneck uses per-channel low/high-frequency gains, cutoff 0.25, maximum gain 0.2, and a one-group GroupNorm followed by a 1×1 ReLU–convolutional feed-forward block; all other convolutional blocks use BatchNorm2d. The augmentation parameters are fixed as follows. Geometric augmentation samples an integer rotation angle uniformly from $[0, 358]$ degrees, applies horizontal flipping with probability 0.5, and applies vertical flipping with probability 0.5. The strong view independently applies contrast, brightness, and sharpness perturbations with probability 0.5 each; each corresponding factor is sampled uniformly from $[1.0, 1.1)$. It also applies Gaussian blur with probability 0.5, using a blur radius sampled uniformly from $[0, 1)$. No random augmentation is used for validation or testing. The weak and strong views are generated from the same patient-level image pair, with the strong branch receiving the additional photometric and blur perturbations. This design encourages the weak branch to provide a more stable semantic anchor while the strong branch acts as a perturbation-aware regularizer. The bidirectional diffusion design is adopted because a single diffusion path can only correct pseudo-labels in one direction, whereas the strong-to-weak path transfers robustness from strongly perturbed predictions back to the cleaner weak branch and the weak-to-label path anchors rectification toward available annotations; together, the two paths reduce confirmation bias and improve semantic alignment under severe label scarcity. After the warm-up stage, the rectified pseudo-label y_r is fed back into the segmentation branch and used to compute a rectification loss consisting of weighted cross-entropy and Dice terms on unlabeled samples, thereby allowing the corrected supervision to directly refine the segmentation network during training.

4.5. Evaluation Details

To comprehensively evaluate the segmentation performance of the proposed T-FDR framework, we adopt four widely used metrics in medical image segmentation: Dice Similarity Coefficient (Dice), Jac Index (Jac), 95% Hausdorff Distance (HD95), and Average Surface Distance (ASD). These metrics evaluate the overlap and boundary consistency between the predicted segmentation $\mathcal{P}$ and the ground truth mask $\mathcal{G}$ from different perspectives:

Dice Similarity Coefficient (Dice). The Dice coefficient measures the overlap between the predicted mask and the ground truth:

$$\text{Dice} = \frac{2|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P}| + |\mathcal{G}|}. \tag{60}$$

It ranges from 0 to 1, where higher values indicate better segmentation accuracy.

Jac Index (Jac). Jac index, also known as the Intersection over Union (IoU), is defined as:

$$\text{Jac} = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P} \cup \mathcal{G}|}. \tag{61}$$

Compared with Dice, the Jac index imposes a stricter penalty on segmentation errors.

95% Hausdorff Distance (HD95). The HD95 measures the maximum distance between $\mathcal{P}$ and $\mathcal{G}$, while mitigating the influence of outliers by considering the 95th percentile of distance.

Let $\partial\mathcal{P}$ and $\partial\mathcal{G}$ denote the boundaries of $\mathcal{P}$ and $\mathcal{G}$, respectively. For each point $p \in \partial\mathcal{P}$, we compute its minimum Euclidean distance to $\partial\mathcal{G}$, and for each point $g \in \partial\mathcal{G}$, we compute its minimum distance to $\partial\mathcal{P}$:

$$\begin{cases} d(p, \partial\mathcal{G}) = \min_{g\in\partial\mathcal{G}} \|p - g\|_2, \\ d(g, \partial\mathcal{P}) = \min_{p\in\partial\mathcal{P}} \|g - p\|_2. \end{cases} \tag{62}$$

The combined distance set $\mathcal{D}$ is then:

$$\mathcal{D} = \{d(p, \partial\mathcal{G}) \mid p \in \partial\mathcal{P}\} \cup \{d(g, \partial\mathcal{P}) \mid g \in \partial\mathcal{G}\}. \tag{63}$$

HD95 is defined as the 95th percentile of $\mathcal{D}$:

$$\mathrm{HD95}(\mathcal{P}, \mathcal{G}) = \mathrm{Percentile}_{95}(\mathcal{D}). \tag{64}$$

Average Surface Distance (ASD). ASD computes the average distance between surface points of the predicted and ground truth masks:

$$\mathrm{ASD} = \frac{1}{|\partial\mathcal{P}| + |\partial\mathcal{G}|} \left(\sum_{x\in\partial\mathcal{P}} \min_{y\in\partial\mathcal{G}} \|x - y\| + \sum_{y\in\partial\mathcal{G}} \min_{x\in\partial\mathcal{P}} \|y - x\| \right). \tag{65}$$

$\partial\mathcal{P}$ and $\partial\mathcal{G}$ denote the boundaries of $\mathcal{P}$ and $\mathcal{G}$. This metric complements HD95 by measuring average boundary deviation, offering a more stable indicator of boundary consistency.

Together, these four metrics provide a comprehensive evaluation of both regional overlap and contour precision. We report average values across test sets to ensure robust and unbiased comparisons with the competing methods listed in this paper.

To assess whether the observed improvements over baseline methods were statistically significant, we further conducted paired statistical analyses on case-level metrics computed from the same test subjects. Since no Gaussian assumption was imposed on the metric distributions, we adopted the paired Wilcoxon signed-rank test. For metrics where higher values indicate better performance (Dice and Jac), a one-sided alternative hypothesis was used to test whether T-FDR achieved larger values; for metrics where lower values are preferred (HD95 and ASD), a one-sided alternative hypothesis was used to test whether T-FDR achieved smaller values. When multiple metrics were tested jointly, exact p-values were adjusted using the Holm-Bonferroni procedure to control the family-wise error rate.

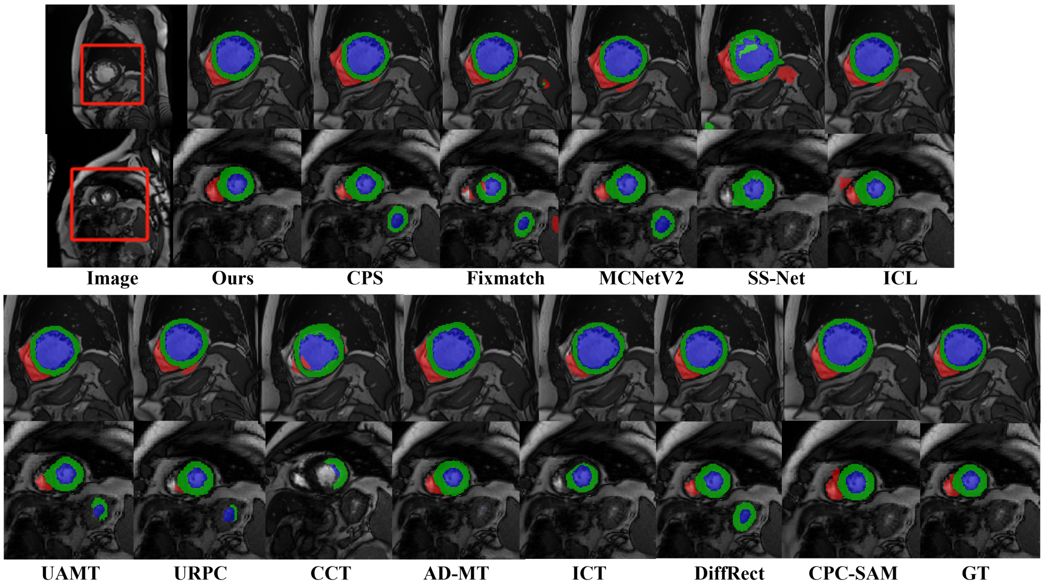

Figure 3: Qualitative segmentation comparisons between T-FDR and competing methods. Red indicates the target region.

4.6. Comparison with State-of-the-art Methods

4.6.1. Results on ACDC Dataset

Table 2 presents the comparative segmentation results of T-FDR against state-of-the-art semi-supervised methods on the ACDC test set under 5% and 10% annotation ratios, including CPC-SAM [27] as a recent transformer- and SAM-based semi-supervised segmentation baseline, with visual comparisons provided in Fig. 3. Following the reviewer suggestion, the table reserves separate columns for the Dice scores of LV, MYO, and RV, while the remaining columns report average test-set metrics over all structures.

Under the 5% labeled setting, T-FDR achieves a Dice coefficient of 89.95 and an HD95 of 3.11 on the test set, outperforming the second-best semi-supervised method DiffRect by approximately 3.0% in Dice and reducing HD95 from 7.18 to 3.11, demonstrating superior boundary delineation under severe label scarcity. Under the 10% labeled setting, T-FDR further achieves a Dice of 90.50 and an HD95 of 2.44, consistently outperforming the compared semi-supervised methods across all four metrics. Visual comparisons in Fig. 3 further confirm that T-FDR produces segmentation boundaries closer to the ground truth, with fewer instances of boundary leakage compared with

Table 2: Structure-wise and average segmentation results on the ACDC test set. LV, MYO, and RV report class-wise Dice scores; the remaining columns report average metrics over all structures. Dice and Jac are reported in percentage (%), while HD95 and ASD are reported in their original units. Each mean and standard deviation is computed independently for the corresponding dataset split and anatomical structure. The best results are highlighted in **bold**.

Method	Labeled ratio	Structure-wise Dice ↑			Dice ↑	Jac ↑	HD95 ↓	ASD ↓
		LV	MYO	RV	Avg.	Avg.	Avg.	Avg.
UA-MT [8]	5%	57.13±26.87	65.72±20.97	80.05±15.99	67.63±23.71	55.29±23.84	21.26±23.35	6.47±9.20
URPC [34]		63.28±25.78	73.18±19.09	84.65±14.32	73.70±22.08	62.41±23.59	10.82±19.15	3.51±8.25
MCNetV2 [35]		69.30±24.76	70.34±21.52	82.32±14.79	73.99±21.60	62.44±22.20	9.73±17.53	4.11±9.74
FixMatch [11]		50.55±21.63	60.98±18.81	76.99±16.81	62.84±36.10	64.12±20.65	11.18±9.86	2.93±2.06
CPS [13]		75.25±18.18	75.46±11.56	83.43±10.70	78.05±14.40	65.91±16.44	19.83±23.47	5.63±6.50
ICT [10]		44.81±30.91	57.09±23.54	67.41±24.50	56.44±28.08	44.13±24.99	25.99±24.87	8.31±9.42
CCT [39]		49.07±26.19	55.48±25.66	68.71±26.83	57.75±27.48	45.45±25.53	30.00±31.62	10.17±12.19
SS-Net [36]		62.53±26.59	70.19±21.83	81.77±16.11	71.50±23.32	59.94±24.14	15.96±23.44	4.80±6.54
AD-MT [37]		78.73±9.53	80.84±5.38	89.54±8.09	83.02±35.28	71.99±16.86	5.18±22.97	1.47±1.32
INCL [38]		77.80±25.69	78.20±20.96	85.92±23.91	80.64±23.52	70.78±18.59	5.29±12.38	1.42±1.25
DiffRect [18]		79.50±12.02	80.10±8.08	87.78±14.89	82.46±11.66	71.76±10.20	7.18±8.59	1.94±1.28
CPC-SAM [27]		88.47±8.47	85.59±9.91	92.29±11.32	88.78±9.90	80.24±9.80	1.40±1.15	0.44±0.36
T-FDR(Ours)		**86.90**±7.12	**88.20**±9.15	**94.75**±12.73	**89.95**±9.67	**82.15**±11.36	**3.11**±4.77	**0.81**±0.58
UA-MT [8]	10%	82.28±6.15	90.97±6.96	84.97±11.58	86.08±9.31	76.66±13.60	15.28±21.64	4.13±5.84
URPC [34]		86.62±8.66	85.44±4.86	91.05±6.47	87.70±7.26	78.79±10.80	5.08±11.45	1.21±2.24
MCNetV2 [35]		87.10±11.41	87.35±6.14	91.80±8.95	88.75±8.83	80.28±7.58	6.16±6.01	1.64±0.61
FixMatch [11]		84.60±9.98	87.10±4.56	92.18±7.09	87.96±6.26	79.37±14.33	5.43±3.22	1.59±1.06
CPS [13]		81.51±13.48	81.40±4.55	88.70±8.46	83.87±10.15	73.39±13.54	8.66±15.24	2.20±3.01
ICT [10]		85.10±10.40	83.54±4.77	89.31±9.34	85.98±8.87	76.38±12.55	6.00±11.80	1.93±3.27
CCT [39]		74.87±12.55	72.07±9.28	82.26±9.80	76.40±11.48	63.12±14.19	16.50±19.27	4.83±5.50
SS-Net [36]		78.52±16.49	78.00±6.89	88.73±8.13	81.75±12.36	70.71±15.32	7.12±13.91	1.75±2.70
AD-MT [37]		80.76±15.96	84.12±18.08	90.93±11.34	85.27±15.13	75.25±7.30	5.33±4.60	1.40±1.24
INCL [38]		86.30±23.53	86.90±16.51	92.84±17.47	88.68±19.17	80.27±5.13	4.34±2.18	1.13±1.01
DiffRect [18]		86.80±3.65	87.40±9.68	93.61±4.34	89.27±5.90	81.13±3.59	3.85±2.01	1.00±0.24
CPC-SAM [27]		88.37±1.25	86.41±2.98	92.49±3.74	89.09±2.57	80.84±4.60	2.35±1.20	0.76±0.55
T-FDR(Ours)		**87.80**±3.53	**88.60**±1.56	**95.10**±3.60	**90.50**±2.90	**82.66**±2.67	**2.44**±0.49	**0.51**±0.68
Fully Supervised [40]	100%	91.92±1.99	96.61±1.19	94.22±1.44	94.25±2.48	89.22±4.37	1.10±0.18	0.32±0.08

competing methods.

As shown in Table 4, the full T-FDR model remains statistically superior to the baseline and key ablated variants on ACDC for Dice and Jac after Holm-Bonferroni correction, while the comparison with DiffRect is not statistically significant under the adopted subject-level paired analysis.

4.6.2. Results on MScmrseg19 Dataset

The quantitative segmentation results on the MScmrseg19 test set are summarized in Table 3. Similar to the ACDC table, we reserve class-wise Dice columns for LV, MYO, and RV and report average Dice, Jac, HD95, and ASD over all structures. As shown in the table, T-FDR achieves a Dice coefficient of 88.51 and an ASD of 1.26, outperforming the compared semi-supervised methods by a margin of 2.0%. Compared with other semi-supervised ap-

Table 3: Structure-wise and average segmentation results on the **MScmrseg19** test set. LV, MYO, and RV report class-wise Dice scores; the remaining columns report average metrics over all structures. Dice and Jac are reported in percentage (%), while HD95 and ASD are reported in their original units. Each mean and standard deviation is computed independently for the corresponding anatomical structure and metric.

Method	Labeled ratio	Structure-wise Dice ↑			Dice ↑	Jac ↑	HD95 ↓	ASD ↓
		LV	MYO	RV	Avg.	Avg.	Avg.	Avg.
UA-MT [8]	20%	$67.77_{\pm 9.07}$	$84.86_{\pm 8.16}$	$71.48_{\pm 16.93}$	$74.70_{\pm 14.11}$	$61.47_{\pm 16.61}$	$39.34_{\pm 55.13}$	$11.28_{\pm 16.80}$
URPC [34]		$83.61_{\pm 5.49}$	$92.19_{\pm 5.19}$	$85.88_{\pm 10.84}$	$87.23_{\pm 8.45}$	$78.27_{\pm 12.46}$	$12.69_{\pm 29.84}$	$3.31_{\pm 5.80}$
MCNetV2 [35]		$77.09_{\pm 8.27}$	$89.54_{\pm 6.55}$	$76.85_{\pm 16.62}$	$81.16_{\pm 12.82}$	$70.06_{\pm 16.55}$	$37.04_{\pm 44.44}$	$10.32_{\pm 12.43}$
FixMatch [11]		$70.65_{\pm 8.52}$	$84.89_{\pm 9.11}$	$66.32_{\pm 23.30}$	$73.95_{\pm 8.20}$	$73.57_{\pm 9.30}$	$17.79_{\pm 5.53}$	$4.81_{\pm 2.33}$
CPS [13]		$80.80_{\pm 6.46}$	$81.40_{\pm 4.28}$	$88.78_{\pm 13.62}$	$83.66_{\pm 8.12}$	$73.03_{\pm 12.14}$	$15.01_{\pm 10.48}$	$4.30_{\pm 2.17}$
ICT [10]		$80.70_{\pm 5.63}$	$81.30_{\pm 4.86}$	$88.98_{\pm 10.33}$	$83.66_{\pm 6.94}$	$73.06_{\pm 9.77}$	$17.24_{\pm 8.22}$	$4.85_{\pm 3.10}$
CCT [39]		$62.88_{\pm 7.31}$	$80.11_{\pm 6.97}$	$70.68_{\pm 11.03}$	$71.22_{\pm 11.14}$	$56.44_{\pm 13.15}$	$63.40_{\pm 45.18}$	$15.29_{\pm 11.01}$
SS-Net [36]		$76.80_{\pm 8.01}$	$88.35_{\pm 7.41}$	$82.53_{\pm 11.37}$	$82.56_{\pm 10.25}$	$71.53_{\pm 14.17}$	$24.40_{\pm 40.67}$	$7.27_{\pm 9.52}$
AD-MT [37]		$78.61_{\pm 5.38}$	$90.37_{\pm 4.16}$	$83.36_{\pm 7.91}$	$84.11_{\pm 11.21}$	$73.33_{\pm 18.78}$	$7.29_{\pm 5.14}$	$2.47_{\pm 1.78}$
INCL [38]		$79.20_{\pm 9.86}$	$90.60_{\pm 14.57}$	$83.19_{\pm 8.14}$	$84.33_{\pm 10.85}$	$73.92_{\pm 12.28}$	$9.95_{\pm 5.45}$	$2.61_{\pm 1.62}$
DiffRect [18]		$82.20_{\pm 5.25}$	$92.10_{\pm 8.55}$	$86.04_{\pm 4.29}$	$86.78_{\pm 6.03}$	$77.13_{\pm 9.83}$	$6.39_{\pm 4.14}$	$1.85_{\pm 0.59}$
CPC-SAM [27]		$83.64_{\pm 9.97}$	$93.51_{\pm 8.34}$	$85.47_{\pm 8.91}$	$87.54_{\pm 9.06}$	$78.86_{\pm 7.10}$	$2.71_{\pm 2.40}$	$1.08_{\pm 0.95}$
T-FDR(Ours)		$\mathbf{84.00}_{\pm 2.46}$	$\mathbf{94.00}_{\pm 4.62}$	$\mathbf{87.53}_{\pm 3.93}$	$\mathbf{88.51}_{\pm 3.67}$	$\mathbf{79.45}_{\pm 5.42}$	$\mathbf{5.17}_{\pm 2.51}$	$\mathbf{1.26}_{\pm 1.08}$
Fully Supervised [40]	100%	$91.92_{\pm 1.99}$	$96.61_{\pm 1.19}$	$94.22_{\pm 1.44}$	$94.25_{\pm 1.29}$	$89.22_{\pm 2.26}$	$0.96_{\pm 0.49}$	$0.18_{\pm 0.17}$

Table 4: Statistical significance analysis on the ACDC test set under the 10% labeled setting. Two-sided paired Wilcoxon signed-rank tests are performed on 20 subject-level paired observations, where ED and ES metrics are first averaged within each subject. For each metric, raw and Holm-Bonferroni-adjusted p-values are reported for the paired comparison between each listed method and T-FDR.

Compared method	Dice			Jac			HD95		
	Mean	Raw p	Adj. p	Mean	Raw p	Adj. p	Mean	Raw p	Adj. p
Baseline	87.1961	1.91×10^{-6}	9.54×10^{-6}	77.2654	1.91×10^{-6}	9.54×10^{-6}	6.7609	1.91×10^{-6}	9.54×10^{-6}
DiffRect	90.3389	0.3884	0.3884	82.6581	0.4091	0.4091	2.5535	0.4265	0.5781
w/o Rectification	88.6021	1.91×10^{-6}	9.54×10^{-6}	79.8345	1.91×10^{-6}	9.54×10^{-6}	5.2417	5.72×10^{-6}	2.29×10^{-5}
w/o latent diffusion loss	89.8254	0.0027	0.0054	81.8118	0.0017	0.0034	3.0933	0.2891	0.5781
w/o refinement components	89.2448	1.05×10^{-4}	3.15×10^{-4}	80.7281	1.34×10^{-4}	4.01×10^{-4}	3.9440	0.0045	0.0135
T-FDR	**90.5958**	–	–	**83.0002**	–	–	**2.4317**	–	–

proaches, T-FDR shows more stable performance across different evaluation metrics.

4.6.3. Results on M&MS Dataset

Table 5 presents the average segmentation results on the M&MS test set, while Table 6 separately reports vendor-wise Dice scores for LV, MYO, and RV under Vendors A–D. This split presentation improves readability and better reflects the multi-center nature of M&MS. Among the compared semi-supervised methods, T-FDR achieves the best overall performance, with Dice, Jac, HD95, and ASD scores of 86.02, 76.23, 2.81, and 0.76 at the 20% labeling ratio. The vendor-wise structure results further show that T-FDR

Table 5: Average segmentation results on the **M&MS** test set. Dice and Jac are reported in percentage (%), while HD95 and ASD are reported in their original units. Each mean and standard deviation is computed independently for the corresponding metric on the M&MS test set.

Method	Labeled ratio	Dice ↑	Jac ↑	HD95 ↓	ASD ↓
UA-MT [8]	20%	76.07±18.21	61.89±16.99	36.51±18.67	11.42±9.24
URPC [34]		74.66±15.79	64.43±14.98	45.88±19.20	15.10±9.93
MCNetV2 [35]		83.59±7.71	73.30±19.48	19.73±6.18	5.77±2.78
FixMatch [11]		55.37±21.55	44.33±20.83	68.27±61.10	42.12±29.73
CPS [13]		71.83±16.08	58.32±23.33	50.46±23.08	16.35±12.22
ICT [10]		80.90±13.84	69.91±14.46	28.87±9.68	8.92±3.56
CCT [39]		43.09±35.53	34.49±28.67	60.83±37.68	38.07±23.91
SS-Net [36]		73.22±16.11	57.18±17.57	42.65±18.73	43.57±14.30
AD-MT [37]		72.27±11.73	62.37±19.98	36.47±18.45	12.91±11.37
INCL [38]		78.23±11.82	69.48±22.58	27.30±5.65	9.73±8.81
DiffRect [18]		83.91±10.28	73.45±13.65	5.41±3.88	1.53±1.19
CPC-SAM [27]		85.23±7.52	77.10±6.10	2.82±8.89	0.82±2.47
T-FDR(Ours)		**86.02**±8.60	**76.23**±10.52	**2.81**±1.76	**0.76**±0.59
Fully Supervised [40]	100%	90.13±3.40	82.59±5.37	1.18±0.67	0.18±0.17

Table 6: Vendor-wise class Dice scores on the **M&MS** test set. Dice is reported in percentage (%). Under each vendor, LV, MYO, and RV denote structure-wise Dice scores, and each mean and standard deviation is computed independently for the corresponding anatomical structure and vendor subset.

Method	Labeled ratio	Dice ↑											
		Vendor A			Vendor B			Vendor C			Vendor D		
		LV	MYO	RV	LV	MYO	RV	LV	MYO	RV	LV	MYO	RV
UA-MT [8]	20%	91.91±4.04	73.15±7.04	85.46±6.00	87.71±8.25	65.53±14.41	78.72±10.69	83.35±9.19	60.36±13.19	77.66±13.55	83.40±14.78	50.75±12.44	74.84±12.76
URPC [34]		91.87±5.70	75.20±7.80	87.32±5.92	89.72±7.59	74.16±7.80	87.32±12.86	85.72±8.23	63.42±13.92	80.05±15.63	81.70±10.33	39.86±15.03	76.46±11.20
MCNetV2 [35]		72.98±10.07	50.34±16.61	56.03±11.98	72.30±14.90	50.27±18.04	57.25±18.36	71.50±14.29	51.17±19.92	55.86±15.46	73.44±12.04	52.80±18.96	55.44±17.54
FixMatch [11]		82.94±4.07	60.63±7.88	57.27±8.90	84.11±7.23	60.01±12.11	42.74±11.92	73.14±8.23	48.61±14.09	34.54±11.05	71.83±10.86	41.57±10.00	27.45±11.78
CPS [13]		83.38±7.93	68.43±6.49	83.31±5.29	86.66±5.65	70.62±8.38	77.47±15.45	76.83±11.46	60.40±11.61	71.99±20.02	80.72±10.07	57.15±9.79	56.56±19.03
ICT [10]		93.76±3.16	80.37±6.28	88.48±6.06	91.96±4.72	77.60±8.74	85.73±9.25	87.66±7.34	70.34±11.39	82.65±16.70	87.15±7.28	59.96±10.68	77.04±14.55
CCT [39]		91.05±4.10	76.63±7.02	87.69±3.62	62.41±30.86	55.97±18.41	40.27±33.93	68.56±24.09	49.17±18.38	53.81±30.96	1.71±8.06	6.26±4.91	1.32±6.61
SS-Net [36]		53.57±14.30	79.93±5.67	76.10±12.55	74.66±12.28	59.97±13.52	74.60±14.48	76.79±12.81	64.27±18.04	78.49±17.67	72.26±15.68	60.72±16.62	75.35±14.37
AD-MT [37]		79.82±17.96	62.68±9.98	79.81±10.45	84.41±8.32	72.63±4.67	76.78±12.65	74.88±10.91	61.80±14.21	70.43±12.52	78.26±9.36	59.87±16.47	65.89±13.34
INCL [38]		88.15±9.54	70.46±13.58	77.97±10.74	89.65±6.50	71.68±10.87	80.11±12.49	88.65±8.28	69.96±8.91	81.68±13.18	85.78±11.92	60.54±18.47	73.80±17.04
DiffRect [18]		88.94±8.12	76.81±8.68	82.69±22.04	90.43±8.37	76.82±9.51	85.41±10.58	90.29±4.87	76.71±5.97	85.30±8.20	90.70±3.78	76.95±5.32	83.91±8.15
CPC-SAM [27]		89.20±7.85	80.50±4.79	89.30±8.80	88.90±9.25	79.80±4.92	88.60±9.80	88.40±6.05	78.90±8.57	88.10±11.08	87.00±6.74	77.24±8.15	86.82±4.24
T-FDR(Ours)		**89.07**±6.90	**79.93**±5.67	**90.63**±3.53	**90.39**±9.03	**78.51**±9.10	**87.60**±9.61	**90.53**±4.93	**78.55**±5.92	**88.47**±5.97	**91.15**±3.98	**78.47**±5.37	**88.94**±6.29
Fully Supervised [40]	100%	96.00±2.62	86.43±5.22	92.42±4.30	95.61±2.54	86.60±4.97	91.10±4.69	94.43±2.07	83.60±4.22	89.95±5.79	93.58±3.13	79.96±3.92	89.63±4.64

provides consistently strong Dice scores across different scanner vendors and cardiac structures, indicating that the proposed trust-guided and frequency-aware refinement improves robustness under cross-vendor domain variation.

4.6.4. Qualitative Results on the ACDC Dataset

Fig. 5 presents qualitative segmentation results produced by AD-MT, UA-MT, UR-PC, FixMatch, MCNetV2, the ground truth, and our T-FDR for Myo on ACDC (5% labeled protocol, first row) and for RV (second row).

It can be observed that the segmentation results generated by T-FDR

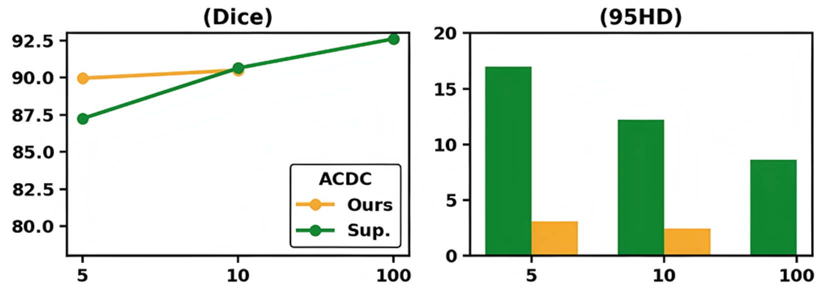

Figure 4: Dice (left) and 95HD (right) comparison between T-FDR(Ours) and fully supervised (Sup.) trained with 5,10 partition protocols and fully labelled data on ACDC dataset.

are visually closer to the ground truth than those of the competing methods. In particular, T-FDR demonstrates superior boundary delineation and structural consistency, whereas other methods exhibit under-segmentation, boundary leakage, or fragmented predictions.

Fig. 4 presents the quantitative comparison between T-FDR and the fully supervised model under different annotation ratios as well as the fully labeled setting on the ACDC dataset. The Dice score of T-FDR consistently improves as the amount of labeled data increases and remains competitive with the fully supervised counterpart. Notably, under limited annotation settings (5% and 10%), T-FDR significantly narrows the performance gap, demonstrating its strong ability to leverage unlabeled data. In terms of HD95, T-FDR achieves substantially lower boundary errors in low-label regimes, indicating that the trust-guided and frequency-aware refinement mechanisms effectively enhance boundary delineation and structural consistency.

4.7. Ablation Experiments

Table 7 presents the ablation experiments we conducted on the ACDC dataset at a 5% annotation ratio, evaluating the contribution of each module (BDCM, TFS, FRRF) to the overall performance. Compared with the baseline, each module consistently improves the segmentation quality, especially on boundary-sensitive indicators. As shown in Table 4, the subject-level paired Wilcoxon signed-rank analysis further confirms that T-FDR significantly outperforms the baseline after Holm-Bonferroni correction.

To further examine the internal contribution of each proposed component, we conduct fine-grained ablation studies on the ACDC dataset under

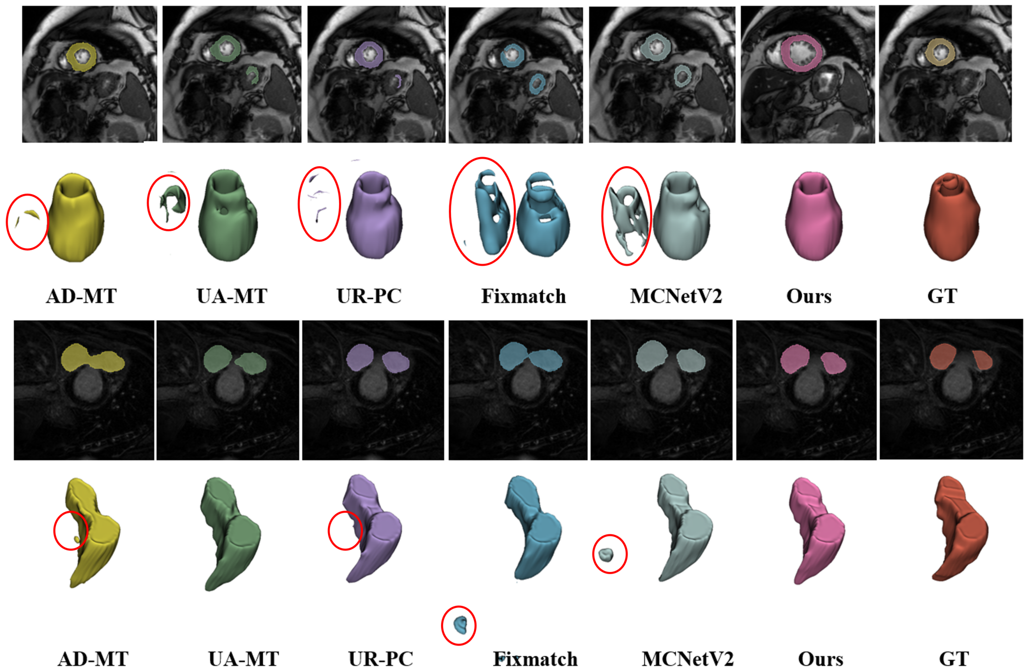

Figure 5: Qualitative segmentation results. Examples from AD-MT, UA-MT, UR-PC, FixMatch, MCNetV2, the ground truth, and our T-FDR for Myo on ACDC (5% labeled protocol, first row) and for RV (second row).

the same low-label setting. Specifically, we analyze the three reliability factors in TFS, the cutoff frequency r_c in FRRF, and the directionality of the diffusion correction in BDCM. These experiments are designed to verify not only whether each module is useful, but also which internal design choices are most responsible for the observed performance gains. For BDCM, we additionally compare the full bidirectional design against strong-to-weak only and weak-to-label only variants under the same 5% labeled setting, backbone, and training budget.

Table 9 quantifies the relative importance of the three reliability factors in TFS. Removing the entropy-based weight slightly degrades performance, indicating that uncertainty-aware suppression is beneficial but not the most critical factor. In contrast, removing the weak–strong consistency weight leads to a clearer drop, especially in HD95, showing that perturbation-induced disagreement contributes substantially to boundary refinement. The

Table 7: Ablation experiments in ACDC segmentation (5% labeled). Dice and Jac are reported in percentage (%), while HD95 and ASD are reported in their original units.

Method	Dice ↑	Jac ↑	HD95 ↓	ASD ↓
Baseline	71.67	68.34	32.92	14.41
+BDCM	85.72	81.79	7.04	2.83
+TFS	81.17	71.06	7.73	6.58
+FRRF	76.87	68.96	12.62	10.57
+BDCM+TFS	88.87	77.25	4.68	2.04
+BDCM+FRRF	88.98	80.87	3.16	0.78
+TFS+FRRF	84.87	75.54	7.03	6.18
+BDCM+TFS+FRRF	89.95	82.15	3.11	0.81

Table 8: Statistical significance analysis between T-FDR and the baseline on the ACDC test set. Paired Wilcoxon signed-rank tests were conducted on 40 paired cases, and exact p-values were adjusted using the Holm-Bonferroni method.

Metric	T-FDR	Baseline	Raw p-value	Adjusted p-value
Dice ↑	89.95	76.52	9.09×10^{-13}	3.64×10^{-12}
Jac ↑	82.15	63.71	9.09×10^{-13}	3.64×10^{-12}
HD95 ↓	3.11	10.99	3.00×10^{-11}	6.00×10^{-11}
ASD ↓	0.81	2.92	5.00×10^{-11}	6.00×10^{-11}

largest degradation is observed when the confidence gate is removed, suggesting that hard confidence filtering is the key mechanism for preventing unreliable pseudo labels from dominating training. Overall, the results confirm that TFS is most effective when complementary reliability cues are combined rather than used in isolation.

Table 10 studies the sensitivity of FRRF to the cutoff frequency r_c. A too-small cutoff ($r_c = 0.10$) pushes excessive spectral content into the high-frequency branch and substantially degrades all metrics, suggesting that the refinement module becomes overly sensitive to noise. In contrast, both $r_c = 0.25$ and $r_c = 0.40$ recover much stronger performance, with $r_c = 0.25$ yielding the best Dice and a well-balanced trade-off across metrics. Although $r_c = 0.40$ slightly improves Jac and achieves a marginally lower HD95, the overall difference is small, so we retain $r_c = 0.25$ as the default setting for stable optimization and balanced boundary enhancement.

Table 9: Fine-grained ablation of TFS components on ACDC. Ent., Cons., and Conf. denote the entropy-based weight, weak–strong consistency weight, and confidence-threshold gate, respectively. Dice and Jac are reported in percentage (%), while HD95 is reported in its original unit.

Variant	Ent.	Cons.	Conf.	Dice ↑	Jac ↑	HD95 ↓
Full TFS	✓	✓	✓	89.95	81.00	3.11
w/o entropy weight	–	✓	✓	89.26	81.37	3.71
w/o consistency weight	✓	–	✓	88.86	80.58	5.30
w/o confidence gate	✓	✓	–	87.23	79.62	6.14

Table 10: Effect of the cutoff frequency r_c in FRRF on ACDC. The cutoff controls the separation between low-frequency structural components and high-frequency boundary components. Dice and Jac are reported in percentage (%), while HD95 is reported in its original unit.

r_c	Dice ↑	Jac ↑	HD95 ↓
0.10	78.36	68.48	7.04
0.25	89.95	81.00	3.11
0.40	89.62	81.70	3.10

Table 11 explicitly compares the full bidirectional BDCM against its two single-direction alternatives under the same 5% labeled ACDC setting. Both single-direction variants substantially outperform removing BDCM entirely, confirming that each diffusion path contributes useful semantic refinement. However, the full bidirectional design still yields the best overall Dice and Jac, indicating that strong-to-weak semantic alignment and weak-to-label structural rectification are complementary rather than redundant.

4.8. Ablation Study of Hyper-parameters

To analyze the sensitivity of the proposed method to the positive confidence threshold γ. We conduct a series of experiments by varying γ from 0 to 1 with an interval of 0.2 in 5% labeled ACDC dataset. For each setting, the corresponding Dice score is recorded and reported in Fig. 6. The results indicate that the segmentation performance gradually improves as γ increases, reaching the best performance when $\gamma = 0.8$, with a Dice score of 89.01 When γ is set too low, a large numb er of low-confidence pseudo labels are involved in training, which introduces noise and degrades performance.

Table 11: Directional ablation of BDCM on ACDC under the 5% labeled setting. All variants use the same backbone, optimization setting, and training budget. Dice and Jac are reported in percentage (%), while HD95 is reported in its original unit.

Variant	Dice ↑	Jac ↑	HD95 ↓
w/o BDCM	76.52	63.70	10.99
Strong-to-weak only	87.16	78.23	4.82
Weak-to-label only	88.92	80.59	4.34
Full bidirectional BDCM	**89.19**	**81.08**	**4.55**

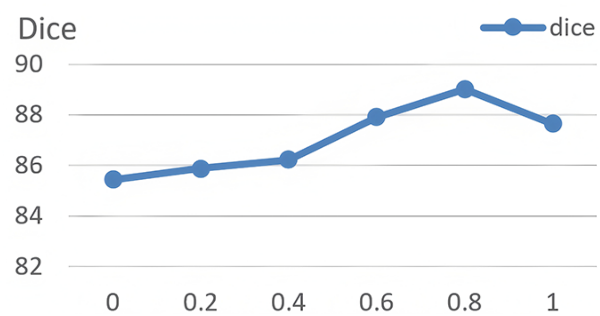

Figure 6: Study of the positive threshold γ.

Conversely, overly large values of γ lead to insufficient supervision signals, resulting in performance degradation. Therefore, $\gamma = 0.8$ is adopted as the default setting in all experiments.

5. Discussion

The experimental results support the central hypothesis of this work: robust semi-supervised cardiac MRI segmentation benefits from jointly modeling pseudo-label reliability, boundary-sensitive feature recovery, and latent semantic correction rather than relying on a single consistency mechanism. Across ACDC, MScmrseg19, and M&MS, T-FDR consistently improves overlap-based metrics while also achieving strong boundary performance, which is particularly important for cardiac MRI because small contour deviations at the endocardial or epicardial boundary can propagate to clinically relevant measurements such as ventricular volume, myocardial mass, and ejection fraction. The improvements observed in HD95 and ASD therefore suggest not only better visual segmentation quality, but also more stable downstream cardiac quantification under limited annotation settings.

The ablation results further clarify how the proposed components answer the initial methodological questions. The trust-guided weighting in TFS reduces the contribution of unreliable pseudo labels, the FRRF module improves boundary sensitivity in uncertain regions, and the BDCM branch refines residual semantic mismatch that remains after conventional weak–strong consistency learning. The fine-grained BDCM analysis is particularly informative: removing diffusion reconstruction, delayed rectification feedback, or auxiliary refinement consistency leads to measurable degradation, indicating that the gains cannot be attributed solely to additional supervision or parameter count. The vendor-wise M&MS results suggest that the method retains useful robustness to the multi-vendor heterogeneity observed in the evaluated data distribution, which is encouraging for realistic deployment scenarios where acquisition characteristics are not fully controlled. Because the current protocol includes the evaluated vendors in the mixed development/evaluation distribution, these results should not be interpreted as evidence of held-out-vendor or held-out-center domain generalization.

At the same time, several limitations should be acknowledged explicitly. First, the diffusion-based correction branch increases computational complexity compared with simpler semi-supervised baselines. The additional refinement network increases training time, memory consumption, and implementation complexity, and its extra latent correction operation also increases inference overhead. This cost is manageable in the present 2D experiments, but it may become more pronounced when scaling to higher-resolution data or volumetric 3D settings. Second, the current framework has been primarily validated on cardiac MRI, and its generalizability to other imaging modalities and anatomical targets remains an open question. Structures in CT, ultrasound, fetal MRI, or abdominal multi-organ imaging may exhibit different texture statistics, boundary sharpness, motion artifacts, and class imbalance patterns, so the relative benefits of trust weighting, frequency refinement, and latent diffusion correction should be re-evaluated rather than assumed to transfer directly.

Third, the present experiments mainly focus on low-label but still non-extreme annotation regimes. While the 5% and 10% protocols on ACDC and the 20% protocols on MScmrseg19 and M&MS already represent challenging semi-supervised settings, performance under extremely scarce annotation budgets such as 1% labeled data remains insufficiently explored. In such cases, the weak branch itself may become unstable, causing pseudo-label noise to increase substantially and reducing the quality of the semantic conditions passed to BDCM. This may limit the reliability of both trust estimation and diffusion-based correction, especially during early training. Fourth, although cardiac MRI often contains low contrast, intensity non-uniformity, motion corruption, partial-volume effects, and vendor-specific artifacts, the current study does not separately quantify sensitivity to each noise or artifact type. The proposed trust and frequency mechanisms are designed to be helpful under uncertain boundary conditions, but systematic stress tests under controlled perturbations would be needed to determine whether the method is equally robust to motion blur, ghosting, bias field distortion, slice misalignment, and scanner-dependent appearance shifts. In addition, the current M&MS experiment follows a mixed-vendor protocol rather than a strict leave-one-vendor-out or leave-one-center-out design. Therefore, it supports robustness to observed multi-vendor heterogeneity but does not establish generalization to an entirely unseen vendor or center; dedicated held-out-domain experiments remain an important direction for future work.

These observations suggest several concrete directions for future research. A first direction is to combine T-FDR with self-supervised or foundation-model pre-training so that the segmentation backbone and semantic correction branch start from stronger anatomical priors under extreme label scarcity. A second direction is to extend the framework from single-modality 2D cardiac MRI to multi-modal or cross-modal settings, such as jointly modeling bSSFP, LGE, and T2 sequences or combining MRI with CT representations. In such settings, trust estimation and latent correction may also be used to align complementary modality-specific cues rather than only weak and strong perturbation views. A third direction is to investigate diffusion correction strategies with lower computational overhead, including adaptive timestep scheduling, selective refinement triggered only in uncertain regions, or lighter denoising networks, without sacrificing semantic correction ability. Finally, extending the current slice-based design to hybrid 2D–3D or fully 3D formulations may better preserve through-plane anatomical continuity and improve boundary consistency across slices, which is especially relevant for volumetric cardiac quantification.

6. Conclusion

In this paper, we proposed **T-FDR**, a semi-supervised medical image segmentation framework that integrates trust-guided feature structuring, frequency-aware latent refinement, and bidirectional diffusion correction. By combining uncertainty-guided supervision, boundary-aware frequency refinement, and latent semantic correction, the proposed framework improves segmentation robustness under limited annotation settings. Experiments on multiple cardiac MRI benchmarks demonstrate that T-FDR achieves strong performance in both overlap and boundary metrics and provides a promising direction for reliable semi-supervised cardiac image segmentation.

CRediT authorship contribution statement

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Chunfeng Yang: Writing - review & editing, Funding acquisition.

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Data availability

The datasets used in this study are publicly available.

Declaration of generative AI use

During the preparation of this work, the authors used ChatGPT to improve language clarity. The authors reviewed and edited the content and take full responsibility for the final manuscript.

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